Minimum norm selection and damping thresholds for critically regularized inertial methods

Abstract

We study inertial gradient and forward–backward methods with extrapolation coefficient $1 - a/k$ and Tikhonov coefficient $b/(sk^2)$. On a real Hilbert space, for $a > 3$, $b > 0$, and $sL < 1$, every orbit converges strongly to the minimum norm minimizer, with objective residual $o(k^{-2})$ and discrete velocity $o(k^{-1})$, without geometric summability or an eventual ball condition. At $a = 3$, targeted Attouch–Czarnecki summability gives strong selection and $O(k^{-2})$, $O(k^{-1})$ rates. Relative flatness permits lower damping for smooth objectives. In finite dimensions with bounded sublevels, growth of order $2 \le q < 4$ only transverse to the affine hull of the minimizer set gives strong selection and both little-o rates for every $a > 1$; quartic transverse growth is also sufficient at $a = 3$. Ambient interior is included as a vacuous-growth case. Explicit coupled examples violate targeted summability, and a shifted quartic example fails relative flatness at every minimizer. A fixed smooth convex Moreau envelope has universal selection exactly for $a \ge 3/2$, with both little-o rates including equality and prescribed equal-unit failures below it. At equality, its corresponding continuous equation fails selection from the same initial position with zero velocity. A separate scalar quartic orbit selects zero while failing the two little-o rates. An elementary coercive example shows that the universal range cannot include $a \le 1$. For positive-semidefinite quadratics, universal convergence is characterized exactly: no spectral condition is needed for $a > 1$; injectivity is necessary and sufficient for $a = 1$; and coercivity is necessary and sufficient for $0 < a < 1$. In the last regime, a noncoercive injective operator admits an unbounded zero-velocity orbit despite having a unique minimizer.

Keywords: convex optimization, inertial gradient method, Tikhonov regularization, minimum norm solution

2020 Mathematics Subject Classification. Primary 90C25; Secondary 65K05, 47J25.

1 Introduction

Let $\mathcal{H}$ be a real Hilbert space. Let $f : \mathcal{H} \to \mathbb{R}$ be convex and continuously Fréchet differentiable, and suppose that its gradient is L-Lipschitz continuous, where $L \ge 0$. We assume that

$$S := \operatorname*{argmin}_{x \in \mathcal{H}} f(x) \neq \varnothing, \qquad x^* := P_S(0), \qquad f_\star := f(x^*).$$

The point x^* is the unique minimum norm element of the closed convex set S. For $a, b, s > 0$, we study the recurrence

$$\begin{aligned} y_k &= x_k + \left(1 - \frac{a}{k}\right)(x_k - x_{k-1}), \\ x_{k+1} &= y_k - s\nabla f(y_k) - \frac{b}{k^2} y_k, \qquad k \ge 1, \end{aligned} \tag{1}$$

with arbitrary initial points $x_0, x_1 \in \mathcal{H}$. The regularization coefficient is $\varepsilon_k = b/(sk^2)$. Its order is critical for the simultaneous study of accelerated objective decay and minimum norm selection.

Vanishing Tikhonov regularization has been studied in inertial forward–backward algorithms by Attouch, Cabot, Chbani, and Riahi [5]. Attouch and László [8] analyzed inertial dynamics and proximal algorithms with a related regularization mechanism. For the explicit recurrence (1), Adly, Ho, and Nguyen [1] obtained objective and velocity estimates of order $O(k^{-2})$ and $O(k^{-1})$ in a critical parameter regime; their analysis does not establish full strong convergence to the minimum norm minimizer in that regime. Their notation is recovered by setting

$$b = sc, \qquad a = \delta\sqrt{sc}, \qquad \varepsilon_k = \frac{c}{k^2}. \tag{2}$$

A closely related critical result is available for accelerated diagonal bilevel methods. Boţ, Chenchene, Csetnek, and Hulett [13, Theorem 4.6] prove convergence of the iterates under an Attouch–Czarnecki summability condition and a subdifferential qualification. For a quadratic norm upper objective, the qualification is automatic, and convergence of its values together with weak convergence yields strong minimum norm convergence. Their work also provides critical selected or ergodic estimates without the same summability condition; see [13, Section 2.3.2]. Theorem 1 concerns the entire sequence in (1), without the summability hypothesis. Section 8 gives a smooth convex objective for which that hypothesis fails.

In continuous time, Li and Xiao [18] recently proved critical minimum norm convergence without an eventual ball condition. The fixed-target energy and signed correction used here follow the same selection principle, but their increments are derived directly from (1).

Theorem 1. *Suppose that $f : \mathcal{H} \to \mathbb{R}$ is convex and continuously Fréchet differentiable, that ∇f is L-Lipschitz continuous, and that $S \neq \varnothing$. Let*

$$a > 3, \qquad b > 0, \qquad s > 0, \qquad sL < 1. \tag{3}$$

Then every sequence generated by (1) *satisfies*

$$\begin{aligned} x_k \longrightarrow x^* \quad \textit{strongly}, \qquad & y_k \longrightarrow x^* \quad \textit{strongly}, \\ f(x_k) - f_\star = o(k^{-2}), \qquad & \|x_k - x_{k-1}\| = o(k^{-1}). \end{aligned} \tag{4}$$

The theorem does not give a quantitative rate for $\|x_k - x^*\|$. The constants in the intermediate estimates depend on the problem, parameters, and initialization. The strict conditions $a > 3$ and $sL < 1$ are used in this proof. The endpoint $a = 3$ is treated below under additional summability or geometric assumptions; the boundary $sL = 1$ is not considered.

The argument also applies to a nonsmooth convex term. For a proper lower semicontinuous convex function g, we write

$$\operatorname{prox}_{sg}(z) := \operatorname*{argmin}_{u\in\mathcal{H}} \left\{ g(u) + \frac{1}{2s}\|u-z\|^2 \right\}.$$

Theorem 2. *Let $f:\mathcal{H}\to\mathbb{R}$ be convex and continuously Fréchet differentiable with L-Lipschitz continuous gradient. Let $g:\mathcal{H}\to\mathbb{R}\cup\{+\infty\}$ be proper, lower semicontinuous, and convex. Suppose that $F=f+g$ has a nonempty minimizer set S, and set $x^*=P_S(0)$ and $F_\star=F(x^*)$. Under (3), start from arbitrary $x_0,x_1\in\mathcal{H}$ and define*

$$\begin{aligned} y_k &= x_k + \left(1-\frac{a}{k}\right)(x_k-x_{k-1}),\\ x_{k+1} &= \operatorname{prox}_{sg}\left(y_k - s\nabla f(y_k) - \frac{b}{k^2}y_k\right), \qquad k\geq 1. \end{aligned} \tag{5}$$

Then $F(x_k)$ is finite for $k\geq 2$, x_k and y_k converge strongly to x^*, and

$$F(x_k)-F_\star = o(k^{-2}), \qquad \|x_k-x_{k-1}\| = o(k^{-1}).$$

The missing damping endpoint can be recovered under a geometric condition in just the normal direction selected by the norm objective. For a set S, write $\sigma_S(p)=\sup_{z\in S}\langle p,z\rangle$, and write H^* for the Fenchel conjugate of a function H.

Theorem 3. *Let f,g,F,S,x^* and $F_\star$ satisfy the structural assumptions of Theorem 2. Let $a=3$, $b,s>0$, and $sL<1$. Put $p=-x^*$ and*

$$D(t) := (F-F_\star)^*(tp) - \sigma_S(tp), \qquad t\geq 0. \tag{6}$$

Suppose that, for some $c>0$ and some integer $N\geq 1$,

$$\sum_{k=N}^{\infty} kD(c/k^2) < \infty. \tag{7}$$

Then every orbit of (5) converges strongly to x^*, as does y_k, and

$$F(x_k)-F_\star = O(k^{-2}), \qquad \|x_k-x_{k-1}\| = O(k^{-1}).$$

The series needs to be finite only on a tail and for one positive scale. Theorem 3 assumes it only for $p=-x^*$, whereas the closest diagonal-method theorem uses a condition for all relevant normal vectors [13, Assumption 2]. Its algorithm assumes $\alpha>3$, and the proof of [13, Theorem 4.6] chooses an auxiliary parameter in $(2,\alpha-1)$, excluding the endpoint considered here. The general $a=3$ case without (7) or further geometric assumptions remains unresolved by the present analysis.

A second way to relax the damping condition is to use relative flatness. The threshold $1+2/\gamma$ and the corresponding unregularized accelerated estimates are established in [4, 10]; unregularized iterate convergence and a linearly corrected objective weight are also given in [3, Chapter 4]. The following theorem adds minimum norm selection for the exact critical Tikhonov recurrence, without a lower growth bound or AC summability.

Theorem 4. *Let f, S, x^* and $f_\star$ satisfy the structural assumptions of Theorem 1. Fix $\gamma > 1$ and consider the inequality*

$$\langle \nabla f(x), x - x^* \rangle \geq \gamma\big(f(x) - f_\star\big). \tag{8}$$

Assume either of the following:

(i) $1 < \gamma \leq 2$ *and* (8) *holds for every* $x \in \mathcal{H}$*;*

(ii) every sublevel set of f is bounded, and for some $\Delta > 0$, (8) *holds for every x with $f(x) \leq f_\star + \Delta$.*

If $b, s > 0$, $sL < 1$, and $a > 1 + 2/\gamma$, then every orbit of (1) *satisfies all conclusions in* (4).

The flatness inequality is relative to the fixed minimum norm point, and no uniqueness is assumed. Local flatness is already part of the classical geometric framework [3, 4, 10]; in (ii), the required entry into the whole low sublevel is proved before selection. This is stronger than assuming the inequality only in a neighborhood of x^*. The global restriction $\gamma \leq 2$ reflects the quadratic upper growth bound from global smoothness; it does not apply to the sublevel condition. Proposition 8.1 gives one fixed smooth nonlinear objective with a nonzero minimum norm point that violates targeted critical AC and satisfies (ii) for every finite $\gamma > 1$. It is therefore covered for every $a > 1$. Proposition 5.1 gives a separate elementary coercive objective for which universal selection holds exactly for $a > 1$.

A different geometric assumption concerns the minimizer set and its affine hull. Interior minimizer sets have long supplied strong convergence and weighted-gradient estimates for inertial dynamics [6, Theorem 3.1] and Hessian-driven dynamics [9, Theorem 2.2]. For a lower-dimensional set, a relative-interior ball controls only the tangential gradient. We combine that estimate with a growth bound in the normal directions, leaving tangential growth unrestricted.

Theorem 5. *Let $f : \mathbb{R}^n \to \mathbb{R}$, $n \geq 1$, be convex and continuously differentiable with globally L-Lipschitz gradient, nonempty $S = \operatorname{argmin} f$, and bounded sublevels. Put $A = \operatorname{aff} S$. Suppose that, for some $c, \Delta > 0$ and $q \in [2, 4]$,*

$$f(x) - f_\star \geq c\, \operatorname{dist}(x, A)^q \qquad \text{whenever } f(x) \leq f_\star + \Delta. \tag{9}$$

For $b, s > 0$ and $sL < 1$, every orbit of (1) *from arbitrary $x_0, x_1 \in \mathbb{R}^n$ satisfies all conclusions in* (4) *in either regime:*

(i) $2 \leq q < 4$ *and* $a > 1$*;*

(ii) $q = 4$ *and* $a = 3$.

The distance in (9) is to aff S, not to S. When S has ambient interior, $A = \mathbb{R}^n$ and the assumption is vacuous, so every $a > 1$ is covered. Relative interior alone gives no such conclusion when A is proper. The zero regularization center is retained even when A is shifted. This theorem is finite-dimensional and smooth; it does not extend the composite theorem to these parameter ranges. No general quartic conclusion throughout $1 < a < 3$ is asserted by this theorem. Proposition 6.1 shows that an extension with the same rates to every $a > 1$ is impossible, even from equal unit initial points. Proposition 6.2 classifies the damping boundary for one fixed smooth convex objective with a compact minimizer segment: universal selection and both little-o rates hold exactly at and above $a = 3/2$, while specified equal-unit initializations fail below it. This is a fixed-objective classification, not a universal quartic theorem. Proposition 6.3 gives a further distinction at equality: the associated continuous equation with the same displayed coefficients fails minimum norm selection from $(0, 1)$ with zero initial velocity. The comparison concerns these two explicit evolutions; the role of gradient and Hessian-driven damping in high-resolution models is already established in [7, 19].

The ravine framework of Davis, Drusvyatskiy, and Jiang [15] also uses normal and tangential decompositions. Their adaptive gradient method assumes prescribed tangential power growth. Here the tangential objective may be flatter than every power, and the conclusion concerns the exact fixed-step inertial recurrence and its selected minimum-norm limit. Proposition 8.2 separates ambient interior from target-relative flatness and AC. Proposition 8.3 gives a genuinely coupled shifted example satisfying quartic transverse growth while failing ordinary relative flatness about every minimizer and targeted AC.

For quadratic objectives the damping range can be classified more precisely. Strong convergence is already known for some unregularized accelerated quadratic methods, even without closed range. Kindermann [17] obtains uniform residual-polynomial bounds for Nesterov regularization with a prescribed inverse-problem initialization. Bartz, Bauschke, Gao, and Moursi [12] prove strong convergence of FISTA in an affine-quadratic setting without a closedness assumption. The following result concerns the time-dependent critical Tikhonov multiplier and arbitrary initial states. Its small-spectrum obstruction can occur even when the initial velocity is zero.

Theorem 6. *Let $\mathcal{H} \neq \{0\}$, let Q be a bounded selfadjoint operator with $0 \leq Q \leq LI$, and let $a, b, s > 0$ with $sL < 1$. For each $x^* \in (\ker Q)^\perp$, consider*

$$f(x) = \frac{1}{2} \langle Q(x - x^*), x - x^* \rangle . \tag{10}$$

The iterates of (1) converge strongly to x^* for every such target and every initial pair if and only if one of the following holds:

1. *$a > 1$;*
2. *$a = 1$ and $\ker Q = \{0\}$;*

3. $0 < a < 1$ *and* $Q \geq \mu I$ *for some* $\mu > 0$.

When $a = 1$*, a particular orbit converges to* x^* *if and only if* $P_{\ker Q} x_1 = 0$*. When* $0 < a < 1$ *and* $0 \in \sigma(Q)$*, there exists* $v \in \mathcal{H}$ *such that the zero-target orbit with* $x_0 = x_1 = v$ *is unbounded.*

The universal quantifier in this theorem is essential: failure of a spectral condition supplies a bad initialization, not failure of every initialization. No quantitative norm-convergence rate is asserted. In particular, an injective noncoercive operator has a unique quadratic minimizer, yet some zero-velocity orbit is unbounded when $0 < a < 1$.

Section 2 proves Theorem 1, first establishing boundedness and then selection. Section 3 proves the proximal extension through a three-point inequality that does not evaluate g at y_k. Section 4 proves the endpoint result under targeted summability, Section 5 proves selection under relative flatness, Section 6 proves selection under transverse growth and gives a quartic rate obstruction, a fixed nonlinear damping classification, and its continuous endpoint comparison, Section 7 proves the spectral classification, and Section 8 compares the critical parameter and geometric assumptions.

2 Proof of Theorem 1

Fix

$$2 < h < a - 1, \qquad d := a - 1 - h > 0, \qquad \rho := 1 - sL > 0. \tag{11}$$

For $k \geq 1$, put

$$\begin{aligned} &v_k := x_k - x_{k-1}, \quad && r_k := x_k - x^*, && \phi_k := \langle x^*, r_k \rangle, \\ &q_k := \left(\frac{k-1}{k}\right)^2, && U_k := hr_k + (k-a)v_k, && g_k := \nabla f(y_k) + \frac{b}{sk^2} y_k. \end{aligned} \tag{12}$$

We use the positive energy

$$E_k := s(k-1)^2(f(x_k) - f_\star) + \frac{b}{2} q_k \|x_k\|^2 + \frac{1}{2}\|U_k\|^2 + \frac{hd}{2}\|r_k\|^2. \tag{13}$$

Every term in E_k is nonnegative. The target x^* is fixed throughout the argument.

The exact energy increment

We first derive a one-step estimate with all regularization terms retained. Abbreviate $x = x_k$, $r = r_k$, $y = y_k$, $g = g_k$, and $v^+ = v_{k+1}$. The recurrence gives

$$v^+ = \left(1 - \frac{a}{k}\right) v_k - sg, \qquad (k-a)v_k = kv^+ + skg.$$

Consequently

$$U_k = hr + kv^+ + skg, \qquad U_{k+1} = hr + (k-d)v^+.$$

Expanding both squared norms yields the exact identity

$$\begin{aligned}\frac{1}{2}\big(\|U_{k+1}\|^2-\|U_k\|^2\big)+\frac{hd}{2}\big(\|r+v^+\|^2-\|r\|^2\big)\\ =-sk\left\langle hr+kv^+,g\right\rangle-\frac{s^2k^2}{2}\|g\|^2-d\left(k-\frac{a-1}{2}\right)\|v^+\|^2.\end{aligned}\tag{14}$$

The position–velocity terms cancel. The last coefficient uses $d+h=a-1$.

Set $\mu_k=b/(sk^2)$ and $F_k(z)=f(z)+\mu_k\|z\|^2/2$. The function F_k is μ_k-strongly convex and has an $(L+\mu_k)$-Lipschitz gradient. Since $x_{k+1}=y-sg$, smoothness and strong convexity give, for every $z\in\mathcal{H}$,

$$F_k(x_{k+1})-F_k(z)\le\langle g,x_{k+1}-z\rangle+\frac{s^2(L+\mu_k)}{2}\|g\|^2-\frac{\mu_k}{2}\|y-z\|^2.\tag{15}$$

Indeed, subtract the strong convexity inequality at y from the smooth descent inequality at $y-sg$, and then use $y=x_{k+1}+sg$.

For $k>h$, apply (15) at $z=x$ and $z=x^*$ with weights $sk(k-h)$ and skh, respectively. The gradient inner product is $sk\langle g,hr+kv^+\rangle$, which cancels the first term in (14). The difference between the regularization coefficients in the two energies is exactly

$$sk^2\big(F_{k+1}(x_{k+1})-F_k(x_{k+1})\big)=-\frac{b(2k+1)}{2(k+1)^2}\|x_{k+1}\|^2.\tag{16}$$

Since $k(k-h)-(k-1)^2=-[(h-2)k+1]$, we obtain

$$\begin{aligned}E_{k+1}-E_k\le&-s[(h-2)k+1](f(x_k)-f_\star)\\&-\frac{b[(h-2)k+1]}{2k^2}\|x_k\|^2+\frac{hb}{2k}\|x^*\|^2\\&-\frac{b(2k+1)}{2(k+1)^2}\|x_{k+1}\|^2-d\left(k-\frac{a-1}{2}\right)\|v_{k+1}\|^2\\&-\frac{s^2k^2}{2}\left(1-sL-\frac{b}{k^2}\right)\|g_k\|^2\\&-\frac{hb}{2k}\|y_k-x^*\|^2-\frac{b(k-h)}{2k}\|y_k-x_k\|^2.\end{aligned}\tag{17}$$

Thus the change from μ_k to μ_{k+1} is included before any asymptotic estimate is used.

A dissipative estimate with explicit constants

Define

$$A:=2b+\frac{8b^2}{d}+\frac{4h^2b^2}{d}+\frac{8h^2b^2}{\rho},\qquad K\ge\max\left\{2,2a,\sqrt{\frac{2b}{\rho}},\frac{2A}{hb}\right\},\tag{18}$$

where K is an integer. We claim that, for $k\ge K$,

$$\begin{aligned}E_{k+1}-E_k\le&-s(h-2)k(f(x_k)-f_\star)-\frac{hb}{2k}\|r_k\|^2\\&-\frac{d}{4}k\|v_{k+1}\|^2-\frac{\rho}{8}s^2k^2\|g_k\|^2-\frac{hb}{k}\phi_k+\frac{A}{k^2}\|x^*\|^2.\end{aligned}\tag{19}$$

To verify this claim, set

$$c_k := \frac{b(2k+1)}{2(k+1)^2}, \qquad e_k := \frac{b(2k^2-1)}{2k^2(k+1)^2}.$$

For $k \geq 1$, $0 < c_k \leq b/k$ and $0 \leq e_k \leq b/k^2$. The position terms in (17), apart from the term involving $y_k - x^*$, satisfy

$$\begin{aligned} &-\frac{b[(h-2)k+1]}{2k^2}\|x\|^2 - c_k\left\|x+v^+\right\|^2 + \frac{hb}{2k}\|x^*\|^2 \\ &\quad = -\frac{hb}{2k}\|r\|^2 - \frac{hb}{k}\phi_k + e_k\|x\|^2 - 2c_k\left\langle x, v^+\right\rangle - c_k\left\|v^+\right\|^2. \end{aligned} \tag{20}$$

Also $y_k - x^* = r + v^+ + sg$, so

$$-\frac{hb}{2k}\|y_k - x^*\|^2 \leq -\frac{hb}{2k}\|r\|^2 + \frac{hb}{k}\|r\|\left\|v^+\right\| + \frac{hbs}{k}\|r\|\|g\|.$$

The three cross terms are bounded by

$$\begin{aligned} 2c_k\|x\|\left\|v^+\right\| &\leq \frac{dk}{4}\left\|v^+\right\|^2 + \frac{4b^2}{dk^3}\|x\|^2, \\ \frac{hb}{k}\|r\|\left\|v^+\right\| &\leq \frac{dk}{4}\left\|v^+\right\|^2 + \frac{h^2b^2}{dk^3}\|r\|^2, \\ \frac{hbs}{k}\|r\|\|g\| &\leq \frac{\rho}{8}s^2k^2\|g\|^2 + \frac{2h^2b^2}{\rho k^4}\|r\|^2. \end{aligned}$$

Using $\|x\|^2 \leq 2\|r\|^2 + 2\|x^*\|^2$, the remaining positive terms are at most

$$\frac{A}{k^2}\big(\|r\|^2 + \|x^*\|^2\big).$$

The conditions in (18) imply

$$d\left(k - \frac{a-1}{2}\right) \geq \frac{3dk}{4}, \quad 1 - sL - \frac{b}{k^2} \geq \frac{\rho}{2}, \quad \frac{A}{k^2} \leq \frac{hb}{2k}.$$

Insert the preceding bounds into (17). Dropping the remaining nonpositive terms proves (19).

Boundedness before selection

Let

$$\begin{aligned} C_1 &:= h^2 + \frac{hd}{2} + b, \\ \eta &:= \min\left\{1, h-2, \frac{d}{32}, \frac{\rho}{64}, \frac{hb}{4C_1}\right\}, \qquad B := hb + \eta b + A. \end{aligned} \tag{21}$$

All these constants are finite, and $\eta > 0$. The exact expression for U_k gives

$$\frac{1}{2}\|U_k\|^2 \leq h^2\|r_k\|^2 + 2k^2\|v_{k+1}\|^2 + 2s^2k^2\|g_k\|^2.$$

Consequently,

$$\begin{aligned} E_k \le sk^2(f(x_k)-f_\star) + C_1\|r_k\|^2 + 2k^2\|v_{k+1}\|^2 \\ + 2s^2k^2\|g_k\|^2 + b\|x^*\|^2. \end{aligned} \tag{22}$$

The signed forcing in (19) obeys

$$-\frac{hb}{k}\phi_k \le \frac{hb}{4k}\|r_k\|^2 + \frac{hb}{k}\|x^*\|^2.$$

After this estimate, the remaining dissipation in (19) dominates η/k times the four non-constant terms on the right of (22). Here we use $k \ge 1$ for the gradient-square comparison. Hence

$$E_{k+1} \le \left(1-\frac{\eta}{k}\right)E_k + \frac{B}{k}\|x^*\|^2, \qquad k \ge K. \tag{23}$$

Since $0 < \eta \le 1$, induction gives

$$0 \le E_k \le M := \max\left\{E_K, \frac{B}{\eta}\|x^*\|^2\right\}, \qquad k \ge K. \tag{24}$$

This establishes boundedness without using a cluster point or a selection property.

In particular, r_k and U_k are bounded, and

$$0 \le f(x_k) - f_\star \le \frac{M}{s(k-1)^2}.$$

Moreover, $(k-a)v_k = U_k - hr_k$ and $k \ge 2a$ show that kv_k is bounded. We have therefore obtained boundedness of x_k and the objective estimate $O(k^{-2})$ before considering minimum norm convergence.

The signed corrected energy

Define

$$H_k := E_k - \frac{b}{2}\|x^*\|^2. \tag{25}$$

This quantity need not be nonnegative. Its exact decomposition is

$$\begin{aligned} H_k = s(k-1)^2(f(x_k)-f_\star) + \frac{1}{2}\|U_k\|^2 + \frac{hd+bq_k}{2}\|r_k\|^2 \\ + bq_k\phi_k - \frac{b}{2}(1-q_k)\|x^*\|^2. \end{aligned} \tag{26}$$

Set

$$C_0 := h^2 + \frac{hd}{2} + \frac{b}{2}, \qquad \theta := \min\left\{1, \frac{h}{2}, h-2, \frac{d}{32}, \frac{\rho}{64}, \frac{hb}{2C_0}\right\} > 0. \tag{27}$$

The same estimate for U_k now gives

$$\begin{aligned} H_k \le sk^2(f(x_k)-f_\star) + C_0\|r_k\|^2 + 2k^2\|v_{k+1}\|^2 \\ + 2s^2k^2\|g_k\|^2 + bq_k\phi_k. \end{aligned} \tag{28}$$

We have retained the signed last term. The last term of (26) that was omitted is nonpositive.

Add $\theta H_k/k$ to (19), and use (28). The choice of θ absorbs its first four terms into the dissipation. Since $0 \le \theta q_k \le h/2$, it follows that

$$\begin{aligned} H_{k+1} &\le \left(1 - \frac{\theta}{k}\right) H_k - \frac{b(h - \theta q_k)}{k}\phi_k + \frac{A}{k^2}\left\|x^*\right\|^2 \\ &\le \left(1 - \frac{\theta}{k}\right) H_k + \frac{hb}{k}\phi_k^- + \frac{A}{k^2}\left\|x^*\right\|^2, \end{aligned} \tag{29}$$

where $\phi_k^- := \max\{-\phi_k, 0\}$. The remaining task is to show that this forcing vanishes.

Weak cluster points and the projection inequality

We claim that

$$\phi_k^- \longrightarrow 0. \tag{30}$$

If this fails, there exist $\epsilon > 0$ and a subsequence with $\phi_{k_j} \le -\epsilon$. Boundedness gives a further subsequence, still denoted by x_{k_j}, that converges weakly to some $z \in \mathcal{H}$. The convex continuous function f is weakly lower semicontinuous, and its values along this subsequence tend to $f_\star$. Thus

$$f(z) \le \liminf_{j\to\infty} f(x_{k_j}) = f_\star,$$

so $z \in S$. Since $x^* = P_S(0)$,

$$\langle x^*, z - x^*\rangle \ge 0.$$

On the other hand, weak convergence gives $\phi_{k_j} \to \langle x^*, z - x^*\rangle$, contradicting $\phi_{k_j} \le -\epsilon$. This proves (30). Only weak subsequential compactness of bounded sequences in a Hilbert space is used.

Convergence of the corrected energy

Put $\alpha_k = hb\phi_k^- + A\left\|x^*\right\|^2/k$. By (30), $\alpha_k \to 0$, and (29) becomes

$$H_{k+1} \le \left(1 - \frac{\theta}{k}\right) H_k + \frac{\alpha_k}{k}.$$

For any $\epsilon > 0$, choose $N \ge K$ so that $\alpha_k \le \theta\epsilon$ for $k \ge N$. Iterating yields

$$H_k - \epsilon \le (H_N - \epsilon) \prod_{j=N}^{k-1} \left(1 - \frac{\theta}{j}\right).$$

The product tends to zero because the harmonic series diverges. This argument is valid even if H_N is negative. It proves $\limsup H_k \le 0$.

Let

$$T_k := s(k-1)^2(f(x_k) - f_\star) + \frac{1}{2}\|U_k\|^2 + \frac{hd + bq_k}{2}\|r_k\|^2 \geq 0.$$

The exact decomposition (26) gives

$$H_k \geq T_k - b\phi_k^- - \frac{b}{2}(1 - q_k)\|x^*\|^2.$$

Thus $\liminf H_k \geq 0$, and $H_k \to 0$. In turn,

$$0 \leq T_k \leq H_k + b\phi_k^- + \frac{b}{2}(1 - q_k)\|x^*\|^2 \longrightarrow 0.$$

Every nonnegative term in T_k tends to zero. Since $hd > 0$, we obtain $r_k \to 0$, $U_k \to 0$, and $(k-1)^2(f(x_k) - f_\star) \to 0$. Finally,

$$kv_k = \frac{k}{k-a}(U_k - hr_k) \longrightarrow 0, \qquad y_k = x_k + \left(1 - \frac{a}{k}\right)v_k \longrightarrow x^*.$$

These are precisely the conclusions of Theorem 1. ∎

3 Proof of Theorem 2

The proximity operator of a proper lower semicontinuous convex function is everywhere defined, and its values belong to the function's effective domain. Hence $x_k \in \operatorname{dom} g$ and $F(x_k) < \infty$ for $k \geq 2$, even if x_0 and x_1 are outside that domain.

Set $\mu_k = b/(sk^2)$ and

$$\Psi_k(z) := f(z) + \frac{\mu_k}{2}\|z\|^2, \qquad G_k := \frac{y_k - x_{k+1}}{s}.$$

The proximal optimality condition is

$$G_k - \nabla\Psi_k(y_k) \in \partial g(x_{k+1}). \tag{31}$$

Fix $z \in \operatorname{dom} g$. The subgradient inequality in (31), the smoothness inequality for Ψ_k at y_k, and its strong convexity inequality at the same point give

$$\begin{aligned} g(x_{k+1}) - g(z) &\leq \langle G_k - \nabla\Psi_k(y_k), x_{k+1} - z\rangle, \\ \Psi_k(x_{k+1}) &\leq \Psi_k(y_k) + \langle \nabla\Psi_k(y_k), x_{k+1} - y_k\rangle + \frac{L + \mu_k}{2}\|x_{k+1} - y_k\|^2, \\ \Psi_k(z) &\geq \Psi_k(y_k) + \langle \nabla\Psi_k(y_k), z - y_k\rangle + \frac{\mu_k}{2}\|z - y_k\|^2. \end{aligned}$$

Adding the first two inequalities and subtracting the third yields

$$\begin{aligned} &F(x_{k+1}) + \frac{\mu_k}{2}\|x_{k+1}\|^2 - F(z) - \frac{\mu_k}{2}\|z\|^2 \\ &\quad\leq \langle G_k, x_{k+1} - z\rangle + \frac{s^2(L + \mu_k)}{2}\|G_k\|^2 - \frac{\mu_k}{2}\|y_k - z\|^2. \end{aligned} \tag{32}$$

No value or subgradient of g at y_k is used.

Choose h, d, ρ as in (11), and use v_k, r_k, q_k, U_k as in (12). For $k \geq 2$, define

$$E_k^F := s(k-1)^2(F(x_k) - F_\star) + \frac{b}{2} q_k \|x_k\|^2 + \frac{1}{2} \|U_k\|^2 + \frac{hd}{2} \|r_k\|^2 . \tag{33}$$

The starting index is important: E_1^F is not defined when $F(x_1) = +\infty$.

The update identity

$$v_{k+1} = \left(1 - \frac{a}{k}\right) v_k - sG_k$$

gives (14) with G_k in place of g_k. For $k > \max\{a, h, 1\}$, both x_k and x^* are in $\operatorname{dom} g$ and the weights $sk(k-h)$ and skh are nonnegative. We may therefore apply (32) at these two points with the same weights used in Section 2. The time-dependent quadratic term has exactly the increment (16). Thus (17) holds with E_k^F, $F(x_k) - F_\star$, and G_k replacing E_k, $f(x_k) - f_\star$, and g_k.

The algebra leading from (17) to (19) uses only the update identity and nonnegativity of the objective residual. It therefore gives the same estimate with these replacements and with the same A and K. Since $K \geq 2$, all potential terms are finite. The comparison in (22)–(24) then bounds E_k^F independently of selection. In particular, x_k is bounded and $F(x_k) - F_\star = O(k^{-2})$.

Define $H_k^F = E_k^F - b \|x^*\|^2 /2$. Its expansion and contraction are (26) and (29), with the same replacements. The function $F = f + g$ is proper lower semicontinuous and convex, hence weakly lower semicontinuous. A weak cluster point z of x_k therefore satisfies

$$F(z) \leq \liminf F(x_k) = F_\star.$$

This proves $z \in S$, including membership in $\operatorname{dom} g$; it does not require that this domain be closed. The projection argument proves $\phi_k^- \to 0$. The signed harmonic comparison and nonnegative decomposition from Section 2 now give $r_k \to 0$, $U_k \to 0$, and $(k-1)^2(F(x_k) - F_\star) \to 0$. The identity $kv_k = k(U_k - hr_k)/(k-a)$ completes the velocity and extrapolated-point conclusions. ∎

4 The endpoint $a = 3$ under targeted summability

We first record why the scale in (7) may be chosen freely. This also controls possible infinite conjugate values away from the origin.

Lemma 1. *Let $D : [0, \infty) \to [0, \infty]$ be convex, with $D(0) = 0$, and finite on some interval $[0, t_0]$, where $t_0 > 0$. The following conditions are equivalent, with every series understood on a sufficiently late tail:*

1. *$\sum_k kD(c/k^2) < \infty$ for one, and hence every, $c > 0$;*
2. *$\sum_k D(C/k) < \infty$ for one, and hence every, $C > 0$;*

3. $\int_0^\tau D(t)t^{-2}\,dt < \infty$ *for one, and hence every sufficiently small,* $\tau > 0$.

The first condition is unchanged if c/k^2 *is replaced by* $c/(k+\beta)^2$ *with* $\beta \geq 0$.

Proof. Convexity and $D(0) = 0$ give $D(u)/u \leq D(v)/v$ for $0 < u < v \leq t_0$. Thus both $D(t)$ and $D(t)/t$ are nondecreasing on this interval. On a sufficiently late tail, both $D(C/x)$ and

$$xD(c/x^2) = \frac{c}{x}\frac{D(c/x^2)}{c/x^2}$$

are nonincreasing. Integral comparison and the changes of variable $t = C/x$ and $t = c/x^2$ give

$$\int_N^\infty D(C/x)\,dx = C\int_0^{C/N} \frac{D(t)}{t^2}\,dt,$$
$$\int_N^\infty xD(c/x^2)\,dx = \frac{c}{2}\int_0^{c/N^2} \frac{D(t)}{t^2}\,dt.$$

Changing an upper limit within the finite interval adds a finite integral. Finally, for $k \geq \max\{1, \beta\}$,

$$D\left(\frac{c}{4k^2}\right) \leq D\left(\frac{c}{(k+\beta)^2}\right) \leq D\left(\frac{c}{k^2}\right).$$

The already proved independence of c yields the shifted assertion. ∎

Proof of Theorem 3. The projection property gives $p = -x^* \in N_S(x^*)$. Hence

$$\sigma_S(tp) = t\,\langle p, x^*\rangle \quad (t \geq 0).$$

Consequently D in (6) is convex, $D \geq 0$, and $D(0) = 0$. The finiteness of the series supplies a positive t_0 with $D(t_0) < \infty$; convexity gives finiteness on $[0, t_0]$. Lemma 1 therefore applies without assuming that the conjugate is finite at arbitrary arguments.

Write $v_k = x_k - x_{k-1}$, $r_k = x_k - x^*$, $\phi_k = \langle x^*, r_k\rangle$, and $G_k = (y_k - x_{k+1})/s$. For $k \geq 2$, define

$$E_k = s(k-1)^2(F(x_k) - F_\star) + \frac{b}{2}\left(\frac{k-1}{k}\right)^2 \|x_k\|^2 + \frac{1}{2}\|2r_k + (k-3)v_k\|^2. \tag{34}$$

This energy is finite and nonnegative. The derivation of (17), using the proximal three-point inequality of Section 3, remains valid at $h = 2$, $d = 0$, $a = 3$. In particular, for $k > 2$ it gives

$$\begin{aligned} E_{k+1} - E_k \leq\ & -s(F(x_k) - F_\star) - \frac{b}{2k^2}\|x_k\|^2 + \frac{b}{k}\|x^*\|^2 \\ & - \frac{b(2k+1)}{2(k+1)^2}\|x_{k+1}\|^2 - \frac{s^2k^2}{2}\left(\rho - \frac{b}{k^2}\right)\|G_k\|^2 \\ & - \frac{b}{k}\|y_k - x^*\|^2 - \frac{b(k-2)}{2k}\|y_k - x_k\|^2, \qquad \rho = 1 - sL > 0. \end{aligned} \tag{35}$$

The two interpolation weights are $sk(k-2)$ and $2sk$, so both are nonnegative. No value $g(y_k)$ is used.

We give an explicit estimate for the lower-order terms. Set $N_k = E_k + \|x^*\|^2$ and

$$C_x = \frac{8}{b}, \qquad C_r = \frac{16}{b} + 2, \qquad C_v = 80 + \frac{512}{b}.$$

For $k \geq 6$, the three terms of (34) imply

$$\|x_k\|^2 \leq C_x N_k, \qquad \|r_k\|^2 \leq C_r N_k, \qquad k^2 \|v_k\|^2 \leq C_v N_k. \tag{36}$$

Indeed, the first two inequalities follow from $((k-1)/k)^2 \geq 1/4$ and $\|r_k\|^2 \leq 2\|x_k\|^2 + 2\|x^*\|^2$. For the third, use $k/(k-3) \leq 2$ and

$$(k-3)v_k = \big(2r_k + (k-3)v_k\big) - 2r_k.$$

Let $X = \|x_k\|$, $R = \|r_k\|$ and $V = \|v_k\|$. Grouping the position terms in (35) around x_k, and using $v_{k+1} = (1-3/k)v_k - sG_k$, bounds their sum by

$$-\frac{2b}{k}(\|r_k\|^2 + \phi_k) + \frac{b}{k^2}X^2 + \frac{2b}{k}(X+R)V + \frac{2bs}{k}X\|G_k\|. \tag{37}$$

For completeness, if $c_k = b(2k+1)/(2(k+1)^2)$ and $\zeta_k = -b(2k^2-1)/(2k^2(k+1)^2)$, the remainder after the leading term is exactly

$$\begin{aligned} &-\zeta_k X^2 - c_k\big(2\langle x_k, v_{k+1}\rangle + \|v_{k+1}\|^2\big) \\ &\quad - \frac{b}{k}\big(2(1-3/k)\langle r_k, v_k\rangle + (1-3/k)^2 V^2\big), \end{aligned}$$

after discarding the last nonpositive term of (35). The bounds $-\zeta_k \leq b/k^2$, $c_k \leq b/k$ and $0 \leq 1-3/k \leq 1$ give (37).

By (36) and Young's inequality, the middle two terms of (37) are at most

$$\frac{b}{k^2}(3C_x + 2C_r + C_v)N_k = \frac{568 + 84b}{k^2}N_k.$$

The last term is at most

$$\frac{\rho}{8}s^2k^2\|G_k\|^2 + \frac{8b^2}{\rho k^4}X^2 \leq \frac{\rho}{8}s^2k^2\|G_k\|^2 + \frac{64b}{\rho k^2}N_k.$$

Thus, with $C_0 = 568 + 84b + 64b/\rho$, for $k \geq \max\{6, \sqrt{2b/\rho}\}$ we obtain

$$\begin{aligned} E_{k+1} - E_k \leq\ & -s(F(x_k) - F_\star) - \frac{2b}{k}\|r_k\|^2 - \frac{2b}{k}\phi_k \\ & - \frac{\rho}{8}s^2k^2\|G_k\|^2 + \frac{C_0}{k^2}(E_k + \|x^*\|^2). \end{aligned} \tag{38}$$

Fenchel's inequality gives, for every sufficiently small $\eta > 0$,

$$D(\eta) \geq -\eta\phi_k - (F(x_k) - F_\star).$$

Take $\eta_k = 4b/(sk)$. On a sufficiently late tail it lies in the finite domain of D, and

$$-\frac{2b}{k}\phi_k \le \frac{s}{2}(F(x_k) - F_\star) + \frac{s}{2}D(\eta_k).$$

Substituting into (38) yields

$$\begin{aligned} E_{k+1} \le \left(1 + \frac{C_0}{k^2}\right) E_k - \frac{s}{2}(F(x_k) - F_\star) - \frac{2b}{k}\left\|r_k\right\|^2 \\ - \frac{\rho}{8}s^2k^2\left\|G_k\right\|^2 + \frac{C_0}{k^2}\left\|x^*\right\|^2 + \frac{s}{2}D(\eta_k). \end{aligned} \tag{39}$$

Lemma 1 makes the last two terms summable. A direct discrete Gronwall argument first bounds E_k: divide by the partial products of $1 + C_0/k^2$, which are bounded above and away from zero. Adding the remaining tail of the normalized additive terms gives a nonincreasing nonnegative sequence. It follows that E_k has a finite limit and

$$\sum_k (F(x_k) - F_\star) < \infty, \qquad \sum_k \frac{\left\|r_k\right\|^2}{k} < \infty, \qquad \sum_k k^2\left\|G_k\right\|^2 < \infty. \tag{40}$$

The bounded energy gives the stated objective and velocity estimates.

It remains to obtain strong convergence. By (36), $\|v_k\| \le M/k$ on a tail for some $M < \infty$. If $M = 0$, the conclusion follows immediately from the middle series in (40). Otherwise, if $\left\|r_{k_j}\right\| \ge \epsilon > 0$ along a subsequence, choose $\theta > 0$ with $M\log(1 + 2\theta) < \epsilon/2$. For all large j, every $n \in [k_j, \lfloor(1 + \theta)k_j\rfloor]$ satisfies

$$\left\|r_n - r_{k_j}\right\| \le M\sum_{i=k_j+1}^{n}\frac{1}{i} < \frac{\epsilon}{2}.$$

Each such block contributes a fixed positive amount to $\sum_k \left\|r_k\right\|^2/k$. An infinite disjoint subcollection contradicts its finiteness. Hence $x_k \to x^*$ strongly; since $v_k \to 0$, also $y_k \to x^*$. ∎

When $x^* = 0$, the function D is identically zero, so the summability condition holds automatically. This is a restriction on the position of the minimizer set: translating a general problem also translates its fixed Tikhonov center. The theorem does not resolve the general endpoint without (7), and it asserts no little-o rate at $a = 3$.

5 Selection under relative flatness

The linearly corrected objective weight below follows the flatness energy used for unregularized inertial methods in [3, 4]; see [3, Chapter 4, equation (4.50)]. We derive its interaction with the critical Tikhonov term explicitly.

Mechanical Lyapunov functions are standard in vanishing-damping regularization; see, for example, [2, Lemma 3.1]. The next estimate records the increment for the present discrete scheme, including the changing regularization coefficient.

Lemma 2. *Let f satisfy the structural assumptions of Theorem 1, and let $a, b, s > 0$ with $sL < 1$. Set*

$$\beta_k = 1 - a/k, \qquad \mu_k = b/(sk^2), \qquad W_k = f(x_k) - f_\star + \frac{\mu_k}{2}\|x_k\|^2 + \frac{1}{2s}\|v_k\|^2 .$$

For every $k \geq 1$,

$$\begin{aligned} W_{k+1} - W_k \leq & -\left(\frac{1-\beta_k^2}{2s} + \frac{\mu_k \beta_k^2}{2}\right)\|v_k\|^2 \\ & - \frac{s}{2}\left(\rho - \frac{b}{k^2}\right)\|g_k\|^2 - \frac{\mu_k - \mu_{k+1}}{2}\|x_{k+1}\|^2 , \qquad \rho = 1 - sL. \end{aligned} \tag{41}$$

If all sublevel sets of f are bounded, then x_k, y_k, v_k are bounded and $f(y_k) \to f_\star$.

Proof. Apply (15) with $z = x_k$ and use $sg_k = \beta_k v_k - v_{k+1}$. The identity

$$\langle g_k, v_{k+1}\rangle + \frac{\|v_{k+1}\|^2 - \|v_k\|^2}{2s} = -\frac{1-\beta_k^2}{2s}\|v_k\|^2 - \frac{s}{2}\|g_k\|^2$$

then gives (41) after adding the parameter change $(\mu_{k+1} - \mu_k)\|x_{k+1}\|^2/2$.

For $k \geq K_W \geq \max\{1, a, \sqrt{2b/\rho}\}$, with K_W an integer, all terms on the right of (41) are nonpositive and

$$W_{k+1} - W_k \leq -\frac{s\rho}{4}\|g_k\|^2 .$$

The nonnegative sequence W_k is therefore bounded on this tail, and $\sum_k \|g_k\|^2 < \infty$. The objective term puts x_k in the sublevel $f \leq f_\star + W_{K_W}$, which is bounded by hypothesis. The kinetic term bounds v_k, and hence also y_k. Since $g_k \to 0$ and $\mu_k y_k \to 0$, we have $\nabla f(y_k) \to 0$. Convexity gives

$$0 \leq f(y_k) - f_\star \leq \langle \nabla f(y_k), y_k - x^*\rangle \leq \|\nabla f(y_k)\|\,\|y_k - x^*\| \longrightarrow 0.$$

This entry argument uses neither selection nor compactness of bounded sets. ∎

Proof of Theorem 4. In case (i), (8) holds at every y_k. In case (ii), Lemma 2 gives an index N such that it holds at every y_k with $k \geq N$. Take $N = 1$ in case (i), and work on this tail. No application of flatness away from these extrapolated points will be needed.

Choose and define

$$\frac{2}{\gamma} < h < a - 1, \qquad d = a - 1 - h > 0, \qquad m = h(\gamma - 1) > 0, \qquad c = h\gamma - 2 > 0. \tag{42}$$

Keep $r_k, v_k, \phi_k, q_k, U_k$ and g_k as in (12), now with this choice of h, and put $\rho = 1 - sL > 0$. The energy is

$$\mathcal{E}_k = s(k-1)(k-1+m)(f(x_k) - f_\star) + \frac{b}{2}q_k\|x_k\|^2 + \frac{1}{2}\|U_k\|^2 + \frac{hd}{2}\|r_k\|^2 . \tag{43}$$

All its terms are nonnegative. Unlike the preceding general argument, h may be less than two.

Write E_k^0 for (13) with the present h, d. The kinetic identity (14) is unchanged. The comparison at $z = x^*$ in (15) is strengthened by

$$-(\gamma - 1)(f(y_k) - f_\star)$$

on its right-hand side, by (8). The comparison at $z = x_k$ remains unchanged. Their weights skh and $sk(k-h)$ are positive on $k \geq 2a$. Thus the right-hand side of (17) gains the term $-smk(f(y_k) - f_\star)$ for the increment of E_k^0.

The regularization estimates leading to (19) require only $h, d, \rho > 0$, not $h > 2$; the objective coefficient is retained with its sign. Adding the increment of $sm(k-1)(f(x_k)-f_\star)$ therefore changes the objective contribution to

$$-s(ck - m)(f(x_k) - f_\star) + smk\big(f(x_{k+1}) - f(y_k)\big).$$

This is the reason for the linear correction in (43). From the actual update and smoothness,

$$f(x_{k+1}) - f(y_k) \leq -s\left(1 - \frac{sL}{2}\right)\|g_k\|^2 + \frac{b}{k^2}\langle y_k, g_k\rangle.$$

The first term is nonpositive. Hence only $(sbm/k)\langle y_k, g_k\rangle$ needs an additional bound.

Use the constant A from (18) with the present h, d, ρ, and set

$$\begin{aligned} C_y &= \frac{16}{b} + 32 + \frac{32h}{d}, \quad & C &= \frac{4b^2m^2C_y}{\rho}, \\ C_1 &= h^2 + \frac{hd}{2} + b, & C_0 &= h^2 + \frac{hd}{2} + \frac{b}{2}. \end{aligned} \tag{44}$$

Choose an integer

$$K_0 \geq \max\left\{N, 2, 2a, \sqrt{\frac{2b}{\rho}}, \frac{2A}{hb}, \frac{2m}{c}\right\}.$$

For $k \geq K_0$, positivity of (43) gives

$$\|x_k\|^2 \leq \frac{8}{b}\mathcal{E}_k, \qquad \|U_k\|^2 \leq 2\mathcal{E}_k, \qquad \|r_k\|^2 \leq \frac{2}{hd}\mathcal{E}_k.$$

Since $y_k = x_k + (U_k - hr_k)/k$, these estimates imply $\|y_k\|^2 \leq C_y\mathcal{E}_k$. Young's inequality now gives

$$\frac{sbm}{k}|\langle y_k, g_k\rangle| \leq \frac{\rho}{16}s^2k^2\|g_k\|^2 + \frac{4b^2m^2}{\rho k^4}\|y_k\|^2 \leq \frac{\rho}{16}s^2k^2\|g_k\|^2 + \frac{C}{k^4}\mathcal{E}_k.$$

Also $ck - m \geq ck/2$. The resulting one-step estimate is

$$\begin{aligned} \mathcal{E}_{k+1} - \mathcal{E}_k \leq\ & -\frac{sc}{2}k(f(x_k) - f_\star) - \frac{hb}{2k}\|r_k\|^2 - \frac{d}{4}k\|v_{k+1}\|^2 \\ & - \frac{\rho}{16}s^2k^2\|g_k\|^2 - \frac{hb}{k}\phi_k + \frac{A}{k^2}\|x^*\|^2 + \frac{C}{k^4}\mathcal{E}_k. \end{aligned} \tag{45}$$

All regularization terms still refer to y_k in the update.

For clarity, explicit constants close the boundedness argument. Let

$$\begin{aligned}\eta &= \min\left\{1, \frac{c}{2(1+m)}, \frac{d}{32}, \frac{\rho}{128}, \frac{hb}{4C_1}\right\}, \qquad B = hb + \eta b + A,\\ \theta &= \min\left\{1, \frac{h}{2}, \frac{c}{2(1+m)}, \frac{d}{32}, \frac{\rho}{128}, \frac{hb}{2C_0}\right\}.\end{aligned} \tag{46}$$

Choose an integer $K \geq \max\{K_0, (2C/\eta)^{1/3}\}$. The identity $U_k = hr_k + kv_{k+1} + skg_k$ and the corrected objective weight yield

$$\begin{aligned}\mathcal{E}_k \leq\ & s(1+m)k^2(f(x_k) - f_\star) + C_1 \|r_k\|^2 + 2k^2 \|v_{k+1}\|^2 \\ & + 2s^2k^2 \|g_k\|^2 + b \|x^*\|^2 .\end{aligned} \tag{47}$$

Estimate $-hb\phi_k/k \leq hb \|r_k\|^2/(4k) + hb \|x^*\|^2/k$ in (45). The remaining loss dominates η/k times the four nonconstant terms of (47). Since $C/k^4 \leq \eta/(2k)$,

$$\mathcal{E}_{k+1} \leq \left(1 - \frac{\eta}{2k}\right)\mathcal{E}_k + \frac{B}{k}\|x^*\|^2 .$$

Consequently

$$0 \leq \mathcal{E}_k \leq M := \max\left\{\mathcal{E}_K, \frac{2B}{\eta}\|x^*\|^2\right\}, \qquad k \geq K. \tag{48}$$

This proves boundedness and the preliminary $O(k^{-2})$ and $O(k^{-1})$ estimates without invoking selection.

Finally let $\mathcal{H}_k = \mathcal{E}_k - b\|x^*\|^2/2$. Its exact decomposition is

$$\begin{aligned}\mathcal{H}_k =\ & s(k-1)(k-1+m)(f(x_k) - f_\star) + \frac{1}{2}\|U_k\|^2 + \frac{hd + bq_k}{2}\|r_k\|^2 \\ & + bq_k\phi_k - \frac{b}{2}(1-q_k)\|x^*\|^2 .\end{aligned} \tag{49}$$

The signed upper bound corresponding to (28) has objective coefficient $s(1+m)k^2$ and the present C_0. Adding $\theta\mathcal{H}_k/k$ to (45), and using (48), gives

$$\mathcal{H}_{k+1} \leq \left(1 - \frac{\theta}{k}\right)\mathcal{H}_k + \frac{hb}{k}\phi_k^- + \frac{A}{k^2}\|x^*\|^2 + \frac{CM}{k^4}. \tag{50}$$

The weak-cluster projection argument of Section 2 applies because boundedness and vanishing objective residual are now established. Hence $\phi_k^- \to 0$. The forcing in (50), after multiplication by k, tends to zero, so the signed harmonic comparison gives $\limsup \mathcal{H}_k \leq 0$. Equation (49) gives the reverse lower limit and forces each of its nonnegative quadratic and objective terms to vanish. Therefore $r_k \to 0$, $U_k \to 0$, and $k^2(f(x_k) - f_\star) \to 0$. As before, $kv_k \to 0$ and $y_k \to x^*$. ∎

The sublevel condition is not replaced by a neighborhood condition at one selected point. The entry lemma proves value convergence, not prior minimum norm selection. The strict finite-exponent boundary $a = 1 + 2/\gamma$ is not covered by this argument.

A coercive example with a sharp lower damping boundary

The following single objective separates the condition $a > 1$ from its boundary. Its targeted AC condition is automatic because $x^* = 0$; it is distinct from the non-AC example of Proposition 8.1.

Proposition 5.1. Let

$$f(x) = \frac{1}{4}\big(\max\{|x| - 1, 0\}\big)^2, \qquad x \in \mathbb{R}, \tag{51}$$

and fix $b = s = 1$ in (1). For every $a > 0$, convergence to the minimum norm minimizer 0 for all initial pairs holds if and only if $a > 1$. For every $0 < a \le 1$, the initialization $x_0 = x_1 = 1$ fails to converge to 0.

Proof. The function is convex and C^1, with $L = 1/2$, bounded sublevel sets and $S = [-1, 1]$. If $r = |x| - 1 > 0$, then

$$\frac{f'(x)x}{f(x)} = 2 + \frac{2}{r}.$$

For every finite $\gamma > 1$, condition (8) therefore holds on $f \le 1/(4\gamma^2)$; inside S both sides vanish. Given $a > 1$, choose a finite $\gamma > \max\{1, 2/(a-1)\}$. Theorem 4(ii) proves the positive assertion for every initialization.

First take $a = 1$. Define the scalar sequence independently by

$$u_0 = u_1 = 1, \qquad z_k = u_k + \left(1 - \frac{1}{k}\right)(u_k - u_{k-1}), \qquad u_{k+1} = q_k z_k, \qquad q_k = 1 - \frac{1}{k^2}.$$

Writing $v_k = u_k - u_{k-1}$, direct calculation gives

$$u_2 = 0, \quad u_3 = -\frac{3}{8}, \quad u_4 = -\frac{5}{9}, \qquad z_1 = 1, \quad z_2 = -\frac{1}{2}, \quad z_3 = -\frac{5}{8}.$$

For $k \ge 4$ set $J_k = u_k^2 + k(k-2)v_k^2$. Expanding one update yields

$$J_{k+1} = q_k u_k^2 + q_k^2 (k-1)^2 v_k^2, \qquad J_4 = \frac{41}{72}. \tag{52}$$

Since $q_k \ge q_{k-1} > 0$ and $q_k^2 \le q_{k-1}$ for $k \ge 4$,

$$q_k J_k \le J_{k+1} \le J_k.$$

The second inequality for q_k reduces to $k^4 - 4k^3 + k^2 + 2k - 1 \ge 0$, which holds for $k \ge 4$. Also $\prod_{j=4}^{k-1} q_j = 3k/(4(k-1)) \ge 3/4$. Hence

$$\frac{41}{96} \le J_k \le \frac{41}{72}, \qquad u_k^2 \le \frac{41}{72}, \qquad z_k^2 \le \left(1 + \frac{(1-1/k)^2}{k(k-2)}\right) J_k \le \frac{41}{64} < 1. \tag{53}$$

Together with the initial values, this proves $|z_k| \le 1$ at every index. Thus $f'(z_k) = 0$, and induction identifies $x_k = u_k$ and $y_k = z_k$ in the actual nonlinear recurrence.

Suppose $u_k \to 0$. From (53), the scaled velocity $w_k = kv_k$ is bounded and its absolute value is eventually bounded away from zero. Its exact recurrence is

$$w_{k+1} = \left(1 - \frac{1}{k^2}\right) w_k - \frac{k+1}{k^2} z_k.$$

Therefore $w_{k+1} - w_k \to 0$. Its sign must eventually be fixed, since a sign change between values bounded away from zero would require a nonvanishing jump. But $u_k - u_{k-1} = w_k/k$ would then have a fixed sign and magnitude at least a positive constant times $1/k$, contradicting boundedness of u_k. Selection fails at $a = 1$.

Now let $0 < a < 1$. For $k \geq 2$, the actual step map

$$T_k(t) = \left(1 - \frac{1}{k^2}\right) t - f'(t)$$

is strongly increasing, since for every $t, u \in \mathbb{R}$,

$$(T_k(t) - T_k(u))(t - u) \geq \left(\frac{3}{4} - \frac{1}{2}\right) (t - u)^2.$$

Its only zero is zero, and $1 - a/k > 0$. An adjacent zero pair would therefore propagate backward to contradict $x_1 = 1$, $x_2 = 0$. If $x_k \to 0$, then $y_k \to 0$, so the actual orbit would eventually satisfy the exact flat recurrence in (128), with a nonzero tail state. The flat-energy argument (129)–(131) in Section 7 proves that no such tail can tend to zero for $0 < a < 1$. This contradiction completes the proof. ∎

The negative assertion concerns the specified initialization, not every initial pair. These nonlinear orbits remain bounded by Lemma 2. At $a = 1$, the objective residual is identically zero along the specified orbit, although minimum norm selection fails.

6 Selection under transverse growth

The interior-ball estimate is classical; see [6, Theorem 3.1] and [9, Theorem 2.2]. In a proper affine hull it controls only the tangential gradient. We retain the normal contribution as a gradient-power error and absorb it using the gradient-square dissipation of the discrete step. The comparison centers approach x^*, while the regularization center stays at zero.

Proof of Theorem 5. Write $F = f - f_\star$, $V = A - A$, and $\rho = 1 - sL > 0$. Lemma 2 bounds x_k, y_k, v_k and gives $F(y_k) \to 0$ and

$$G_k := \nabla f(y_k) + \frac{b}{sk^2} y_k \longrightarrow 0.$$

Finite-dimensional compactness, followed by $x_{k+1} - y_k = -sG_k \to 0$, yields

$$\operatorname{dist}(y_k, S) \to 0, \qquad \operatorname{dist}(x_k, S) \to 0. \tag{54}$$

The low-sublevel hypothesis is now available before selection. Put

$$B_y := \sup_k \|y_k\| < \infty.$$

Relative geometry with a gradient-power error

Put $g_k = \nabla f(y_k)$ and $\delta_k = \operatorname{dist}(P_A y_k, S)$. Orthogonal decomposition and convexity at $P_S y_k = P_S P_A y_k$ give

$$F(y_k) \le \delta_k \|P_V g_k\| + \operatorname{dist}(y_k, A) \|P_{V^\perp} g_k\| .$$

Here $\delta_k \to 0$ by (54). The transverse growth assumption and Young's inequality imply

$$\begin{aligned} F(y_k) &\le 2\delta_k \|P_V g_k\| + B_q \|g_k\|^p , \\ p &= \frac{q}{q-1}, \qquad B_q = \frac{2}{p}\left(\frac{2}{cq}\right)^{1/(q-1)} . \end{aligned} \tag{55}$$

Indeed the normal product is at most $c^{-1/q} F(y_k)^{1/q} \|g_k\|$, which is bounded by $F(y_k)/2 + (B_q/2)\|g_k\|^p$.

Choose $z_0 \in \operatorname{ri} S$ and $r_0 > 0$ such that $z_0 + r_0 \overline{B}_V \subset S$, where $\overline{B}_V$ is the closed unit ball of V. For $0 < t \le 1$ set

$$z_t = (1-t)x^* + t z_0, \qquad r_t = t r_0, \qquad \Delta_t = \|z_t - x^*\| .$$

Convexity gives $z_t + r_t \overline{B}_V \subset S$. Relative-ball comparison gives, for every fixed t,

$$\langle g_k, y_k - z_t \rangle \ge F(y_k) + r_t \|P_V g_k\| . \tag{56}$$

If $V = \{0\}$, the relative balls are singletons and the projected gradient is zero. Fix any $\gamma > 1$. On a tail with $2(\gamma-1)\delta_k \le r_t$, equations (55)–(56) give

$$\langle g_k, y_k - z_t \rangle \ge \gamma F(y_k) - D \|g_k\|^p , \qquad D = (\gamma - 1) B_q . \tag{57}$$

The coefficient D is independent of the center and its relative radius. Only the entry index may depend on t. Ordinary relative flatness is not being asserted.

The discrete energy and the additional error

For the moment fix

$$0 < h < a - 1, \quad d = a - 1 - h, \quad m = h(\gamma - 1) > 0, \quad \zeta = h\gamma - 2 > 0. \tag{58}$$

At a fixed center $z = z_t$ write

$$r_k^z = x_k - z, \quad \phi_k^z = \langle z, x_k - z \rangle , \quad U_k^z = h(x_k - z) + (k - a) v_k, \quad \omega_k = \left(\frac{k-1}{k}\right)^2 ,$$

and use the nonnegative energy

$$\mathcal{E}_k^z = s(k-1)(k-1+m)F(x_k) + \frac{b}{2}\omega_k \|x_k\|^2 + \frac{1}{2}\|U_k^z\|^2 + \frac{hd}{2}\|r_k^z\|^2 . \tag{59}$$

The absolute-position term is unchanged. The kinetic identity (14) and the regularization estimates through (19) require only $h, d > 0$, with the objective coefficient retained with its sign. The comparison at z is now strengthened by $(\gamma-1)F(y_k)-D\left\|g_k\right\|^p$ and has weight shk; the comparison at x_k is unchanged. Adding the linear objective-weight correction therefore gives

$$\begin{aligned}\mathcal{E}_{k+1}^z - \mathcal{E}_k^z \le & -s(\zeta k - m)F(x_k) - \frac{hb}{2k}\left\|r_k^z\right\|^2 - \frac{d}{4}k\left\|v_{k+1}\right\|^2 \\ & - \frac{\rho s^2}{8}k^2\left\|G_k\right\|^2 - \frac{hb}{k}\phi_k^z + \frac{A_0}{k^2}\left\|z\right\|^2 \\ & + smk\big(F(x_{k+1}) - F(y_k)\big) + shDk\left\|g_k\right\|^p,\end{aligned} \tag{60}$$

for all sufficiently large k, where

$$A_0 = 2b + \frac{8b^2}{d} + \frac{4h^2b^2}{d} + \frac{8h^2b^2}{\rho}.$$

The tail includes $k \ge \max\{2, 2a, \sqrt{2b/\rho}, 2A_0/(hb)\}$ and the validity of (57). Thus the new defect is included in the increment of the actual recurrence.

Let $\sigma = 1 - sL/2 > 0$. Smoothness along $x_{k+1} = y_k - sG_k$ gives

$$F(x_{k+1}) - F(y_k) \le -s\sigma\left\|G_k\right\|^2 + \frac{b}{k^2}\left\langle y_k, G_k\right\rangle \le \frac{b^2B_y^2}{4s\sigma k^4}.$$

Its contribution to (60) is at most C_d/k^3, with $C_d = mb^2B_y^2/(4\sigma)$.

Put

$$Q = 2^{p-1}shD, \qquad \epsilon_g = \frac{\rho s^2}{16}.$$

The relation $g_k = G_k - by_k/(sk^2)$ implies

$$shDk\left\|g_k\right\|^p \le Qk\left\|G_k\right\|^p + Q\left(\frac{bB_y}{s}\right)^p k^{1-2p}. \tag{61}$$

For $q = 2$, we have $p = 2$, and the first term is absorbed by $\epsilon_g k^2\left\|G_k\right\|^2$ once $k \ge Q/\epsilon_g$. For $2 < q \le 4$, exact scalar maximization gives

$$\begin{aligned}Qku^p - \epsilon_g k^2u^2 &\le C_pk^{-\nu} \qquad (u \ge 0), \\ \nu &= \frac{2(p-1)}{2-p} = \frac{2}{q-2}, \\ C_p &= \frac{2-p}{2}Q\left(\frac{pQ}{2\epsilon_g}\right)^{p/(2-p)}.\end{aligned} \tag{62}$$

The maximum occurs at $u = (pQ/(2\epsilon_g k))^{1/(2-p)}$. Taking also $k \ge 2m/\zeta$ yields

$$\begin{aligned}\mathcal{E}_{k+1}^z - \mathcal{E}_k^z \le & -\frac{s\zeta}{2}kF(x_k) - \frac{hb}{2k}\left\|r_k^z\right\|^2 - \frac{d}{4}k\left\|v_{k+1}\right\|^2 \\ & - \frac{\rho s^2}{16}k^2\left\|G_k\right\|^2 - \frac{hb}{k}\phi_k^z + R_k(z),\end{aligned} \tag{63}$$

where

$$R_k(z) = \frac{A_0 \|z\|^2}{k^2} + \frac{C_d}{k^3} + Q\left(\frac{bB_y}{s}\right)^p k^{1-2p} + C_p k^{-\nu}. \tag{64}$$

The last term is omitted when $q = 2$. All coefficients except the displayed $\|z\|^2$ are center-independent. For $q < 4$, $kR_k(z) \to 0$ uniformly on the segment of centers. For $q = 4$, $\nu = 1$ and the remaining harmonic coefficient is C_p.

A comparison valid for both growth regimes

Set $C_1 = h^2 + hd/2 + b$ and $C_0 = h^2 + hd/2 + b/2$, and choose

$$\begin{aligned} \eta &= \min\left\{1, \frac{\zeta}{2(1+m)}, \frac{d}{32}, \frac{\rho}{128}, \frac{hb}{4C_1}\right\}, \\ \theta &= \min\left\{1, \frac{h}{2}, \frac{\zeta}{2(1+m)}, \frac{d}{32}, \frac{\rho}{128}, \frac{hb}{2C_0}\right\}. \end{aligned} \tag{65}$$

As in (47), the exact identity $U_k^z = hr_k^z + kv_{k+1} + skG_k$ gives

$$\mathcal{E}_k^z \le s(1+m)k^2 F(x_k) + C_1 \|r_k^z\|^2 + 2k^2 \|v_{k+1}\|^2 + 2s^2k^2 \|G_k\|^2 + b\|z\|^2.$$

Using $-hb\phi_k^z/k \le hb\|r_k^z\|^2/(4k) + hb\|z\|^2/k$ in (63) proves

$$\mathcal{E}_{k+1}^z \le \left(1 - \frac{\eta}{k}\right)\mathcal{E}_k^z + \frac{hb + \eta b}{k}\|z\|^2 + R_k(z).$$

Since $R_k(z) = O(k^{-1})$, this bounds $\mathcal{E}_k^z$ for each fixed choice of parameters and center. The bound and the starting index need not be uniform as the center or parameters change.

For $\mathcal{H}_k^z = \mathcal{E}_k^z - b\|z\|^2/2$, retaining the signed pairing in the corresponding upper estimate gives

$$\mathcal{H}_{k+1}^z \le \left(1 - \frac{\theta}{k}\right)\mathcal{H}_k^z + \frac{hb}{k}(\phi_k^z)^- + R_k(z). \tag{66}$$

Let $B = \sup_k \|x_k - x^*\| + \|x^*\| + \|z_0\| < \infty$. Projection normality is used only at x^*:

$$\langle x^*, x_k - x^*\rangle \ge -\|x^*\| \operatorname{dist}(x_k, S).$$

Expanding $\phi_k^{z_t}$ about x^* therefore gives

$$\limsup_k (\phi_k^{z_t})^- \le B\Delta_t. \tag{67}$$

If $\limsup_k kR_k(z_t) \le M$, with M independent of t, the signed harmonic comparison in (66) yields

$$\limsup_k \mathcal{H}_k^{z_t} \le \frac{hbB\Delta_t + M}{\theta}.$$

This follows by subtracting any constant strictly above the right side and iterating the product of $1 - \theta/k$; no summability of the sign defect is needed.

The exact positive decomposition is

$$\begin{aligned} T_k^z &:= s(k-1)(k-1+m)F(x_k) + \frac{1}{2}\left\|U_k^z\right\|^2 + \frac{hd+b\omega_k}{2}\left\|x_k - z\right\|^2 \\ &= \mathcal{H}_k^z - b\omega_k\phi_k^z + \frac{b}{2}(1-\omega_k)\left\|z\right\|^2 \geq 0. \end{aligned} \tag{68}$$

Consequently

$$\limsup_k T_k^{z_t} \leq b\left(1+\frac{h}{\theta}\right)B\Delta_t + \frac{M}{\theta}. \tag{69}$$

Subquartic growth

For $2 \leq q < 4$ and $a > 1$, take

$$h = \min\{1, (a-1)/2\}, \qquad \gamma = 3/h, \qquad m = 3-h, \qquad \zeta = 1.$$

Then $d > 0$ and $M = 0$ in (69). First let $k \to \infty$ for each fixed t, and then let $t \downarrow 0$. The objective coefficient is asymptotic to sk^2, while the state coefficient tends to $(hd+b)/2 > 0$. Thus $k^2F(x_k) \to 0$ and $x_k \to x^*$. Since $U_k^{x^*} = U_k^{z_t} + h(z_t - x^*)$, the same estimate gives $U_k^{x^*} \to 0$. Finally

$$kv_k = \frac{k}{k-a}\big(U_k^{x^*} - h(x_k - x^*)\big) \tag{70}$$

proves $kv_k \to 0$, and $y_k \to x^*$ follows.

Quartic growth at $a = 3$

The harmonic error in the quartic case requires another limit. Keep the actual recurrence parameters $a = 3, b, s$ fixed. For an auxiliary $\varepsilon \in (0, 1/2]$, choose

$$h = 2-\varepsilon, \quad d = \varepsilon, \quad \gamma = \frac{2+\varepsilon}{2-\varepsilon}, \quad m = 2\varepsilon, \quad \zeta = \varepsilon. \tag{71}$$

Now $p = 4/3$ and

$$Q = 2^{4/3}sB_4\varepsilon =: Q_0\varepsilon.$$

Equation (62) gives the exact leading coefficient

$$M = C_*\varepsilon^3, \qquad C_* = \frac{4Q_0^3}{27\epsilon_g^2}. \tag{72}$$

The other terms of $R_k(z)$ are $o(k^{-1})$ for each fixed ε. Their coefficients may grow as $\varepsilon \downarrow 0$; the time limit is taken first.

The contraction constant in (65) obeys

$$\theta \geq \kappa\varepsilon, \qquad \kappa = \min\left\{\frac{1}{32}, \frac{\rho}{128}, \frac{b}{8+b}\right\} > 0. \tag{73}$$

Indeed $\zeta/[2(1+m)] \geq \varepsilon/4$, $d/32 = \varepsilon/32$, $h \geq 3/2$, and $C_0 = 4-3\varepsilon+\varepsilon^2/2+b/2 \leq 4+b/2$. Each remaining entry of the minimum is at least $\kappa\varepsilon$.

For each fixed ε, let $k \to \infty$ and then $t \downarrow 0$ in (69). Using (68) and the triangle inequality for $U_k^{x^*}$ gives

$$\begin{aligned} \limsup_k k^2 F(x_k) &\leq \frac{C_* \varepsilon^3}{s\theta}, \\ \limsup_k \|x_k - x^*\|^2 &\leq \frac{2C_* \varepsilon^3}{b\theta}, \\ \limsup_k \left\|U_k^{x^*}\right\|^2 &\leq \frac{2C_* \varepsilon^3}{\theta}. \end{aligned} \tag{74}$$

For the state estimate we use $hd + b\omega_k \to hd + b \geq b$, not the vanishing coefficient hd. By (73), every right side is $O(\varepsilon^2)$. Letting $\varepsilon \downarrow 0$ proves the state and objective conclusions. For each fixed ε, identity (70), followed by the last bound in (74) and then $\varepsilon \downarrow 0$, proves $kv_k \to 0$. Extrapolation proves $y_k \to x^*$.

The order of limits is time, comparison center, and finally the auxiliary energy parameter. Neither a uniform entry index nor a uniform pre-asymptotic energy bound in these parameters has been assumed. ∎

If S has ambient interior, then $A = \mathbb{R}^n$ and the transverse lower bound is vacuous. The theorem therefore includes strong selection and both little-o rates for every $a > 1$ in that class. Proposition 5.1 shows that its universal damping range cannot include $a \leq 1$. Relative interior alone does not imply the transverse growth assumption when A is proper.

A quartic obstruction to the low-damping rates

The quartic case cannot be added to Theorem 5(i) with the same little-o conclusions for every $a > 1$. This is a rate obstruction, even when the iterates select the minimum norm minimizer and the initial velocity is zero.

Proposition 6.1. Fix

$$K = 10^8, \qquad R = 22^K, \qquad a = \frac{5}{4}, \qquad b = 1, \qquad s = \frac{1}{5},$$

and define the single objective

$$\varphi_R(x) = \begin{cases} R^2 x^4/4, & |x| \leq R^{-1}, \\ 3x^2/2 - 2|x|/R + 3/(4R^2), & |x| \geq R^{-1}. \end{cases} \tag{75}$$

It is convex C^2, has globally 3-Lipschitz derivative and bounded sublevels, and has the unique minimizer zero. The orbit of (1) with $f = \varphi_R$ and $x_0 = x_1 = 1$ satisfies $x_k, y_k \to 0$, but

$$0 < \limsup_k k^2 \varphi_R(x_k) < \infty, \qquad 0 < \limsup_k k|x_k - x_{k-1}| < \infty. \tag{76}$$

The same conclusions hold for the vector orbit of

$$\mathcal{F}_R(u,n) = \frac{1}{4}\operatorname{dist}(u,[-1,1])^2 + \varphi_R(n), \qquad (u_0,n_0) = (u_1,n_1) = (0,1).$$

Here $S = [-1,1]\times\{0\}$ and $A = \mathbb{R}\times\{0\}$; on $\mathcal{F}_R \le 1/(4R^2)$, $\mathcal{F}_R \ge (R^2/4)\operatorname{dist}(\,\cdot\,,A)^4$.

Proof. Let $\varphi = \varphi_1$. Values and the first two derivatives match at ±1, and $0 \le \varphi'' \le 3$. The identity $\varphi_R(x) = R^{-2}\varphi(Rx)$ proves all stated regularity properties. The exact scaling $z_k = Rx_k$ conjugates the orbit to the recurrence for φ with $z_0 = z_1 = R$ and the same a, b, s. We prove the assertions for z_k and scale back at the end.

Lemma 2 first gives boundedness, vanishing objective residual, and vanishing regularized gradient. Uniqueness of the scalar minimizer implies $z_k \to 0$ and $y_k \to 0$. We also need bounded scaled states. The bounded-energy argument preceding (66) applies with

$$q = 4, \qquad h = d = \frac{1}{8}, \qquad \gamma = 24, \qquad m = \frac{23}{8}, \qquad z = 0.$$

It requires only $R_k(0) = O(k^{-1})$, not the final auxiliary-parameter limit at $a = 3$. Thus $k^2\varphi(z_k)$ and $k^2(z_k - z_{k-1})^2$ are bounded. Since z_k eventually lies in the quartic part, the states

$$X_k = \sqrt{k}\, z_k, \qquad V_k = k(z_k - z_{k-1}) \tag{77}$$

are bounded. This step establishes only upper rates.

Define, at every index,

$$E(X,V) = \frac{V^2}{2} + \frac{X^4}{20}, \qquad J_k = E(X_k,V_k) + \frac{5}{12\sqrt{k}}X_kV_k. \tag{78}$$

We derive a finite-step bound before proving that this corrected energy stays away from zero.

The local increment. Put $e = k^{-1/2}$, $\beta = 1 - 5e^2/4$, $Y = X + \beta eV$, and $r = \sqrt{1+e^2}$. Whenever $|eY| \le 1$, the exact scaled update is

$$X_+ = r(Y - e^2Y^3/5 - e^4Y), \qquad V_+ = (1+e^2)(\beta V - eY^3/5 - e^3Y). \tag{79}$$

All regularization terms are retained. We claim that on

$$\frac{1}{20000} \le E \le 5, \qquad 0 < e \le 10^{-4},$$

this update satisfies

$$\begin{aligned} J_{k+1} - J_k &= \frac{1}{k}\left(\frac{E}{3} - \frac{3}{10}X^2V^2 - \frac{1}{50}X^6\right) + \mathcal{R}_k, \\ |\mathcal{R}_k| &\le 160Ek^{-3/2}. \end{aligned} \tag{80}$$

The domain implies $|X|, |V| \le \sqrt{10}$ and hence $|eY| < 1$, so the quartic formula used in the claim is valid.

Here are explicit bounds for the calculation. Set

$$A_0 = X/2 - X^3/5, \quad B_0 = (-1/4 - 3X^2/5)V, \quad U_0 = -X^3/5.$$

Expansion of (79), using $|r - 1 - e^2/2| \le e^4/8$, gives

$$\Delta X = eV + e^2 A_0 + r_X, \qquad \Delta V = eU_0 + e^2 B_0 + r_V,$$

where

$$\begin{aligned} |r_X| &\le e^3 \left(7|V| + \frac{|X|}{1000} \right), \\ |r_V| &\le e^3 \left(\frac{201}{1000}|X|^3 + \frac{601}{1000}|X|V^2 + \frac{1001}{1000}|X| + \frac{3}{5000}|V| \right). \end{aligned} \tag{81}$$

These estimates follow by expanding $Y^3 = (X + \beta eV)^3$ and bounding the remaining powers by $e \le 10^{-4}$ and $X^2, V^2 \le 10$. Useful moment bounds on this domain are

$$V^2 \le 2E, \quad X^4 \le 20E, \quad X^6 \le 200E, \quad X^2 \le 633E, \quad |XV| \le 36E,$$
$$|X^3V|,\ |XV^3|,\ X^2V^2,\ V^4 \le 20E.$$

For example $(|XV|/E)^4 \le 80/E < 36^4$, $(|X^3V|/E)^4 \le 32000E \le 20^4$, and $X^4/E^2 \le 400000 < 633^2$.

The exact polynomial identity

$$\Delta E = V\Delta V + \frac{(\Delta V)^2}{2} + \frac{X^3}{5}\Delta X + \frac{3X^2}{10}(\Delta X)^2 + \frac{X}{5}(\Delta X)^3 + \frac{1}{20}(\Delta X)^4$$

has zero coefficient at order e and coefficient $D = X^4/10 - V^2/4 - 3X^2V^2/10 - X^6/50$ at order e^2. Using (81) and the moment bounds, the six remainder contributions

$$Vr_V, \quad X^3 r_X/5, \quad ((\Delta V)^2 - e^2U_0^2)/2,$$
$$3X^2((\Delta X)^2 - e^2V^2)/10, \quad X(\Delta X)^3/5, \quad (\Delta X)^4/20$$

have absolute values at most $53, 29, 26, 19, 5, 1$ times Ee^3, respectively. Thus $|\Delta E - e^2 D| \le 133Ee^3$. The product expansion also gives

$$\Delta(XV) = e(V^2 - X^4/5) + e^2 XV(1/4 - X^2) + r_P, \qquad |r_P| \le 700Ee^3.$$

Since $|XV(1/4 - X^2)| \le 29E$, it follows that $|\Delta(XV) - e(V^2 - X^4/5)| \le 30Ee^2$ and $|X_+V_+| \le 37E$. Finally $e_+ = e/r$ obeys $|e_+ - e| \le e^3/2$. Inserting these bounds into (78) proves (80), because

$$133 + \frac{5}{12}\,30 + \frac{5}{24}\,37 < 160.$$

This is a finite-step estimate, not a continuous-time approximation.

An energy barrier. Put $\delta = 10^{-4}$. If $\delta/2 \le E \le 4\delta$, then

$$\frac{3}{10}X^2V^2 + \frac{1}{50}X^6 \le \frac{10}{\sqrt{5}}E^{3/2} \le \frac{E}{10}.$$

The last inequality uses $\sqrt{E} \le 1/50$ and $1/\sqrt{5} < 1/2$. Equation (80), valid for every $k \ge K$, therefore gives

$$J_{k+1} - J_k \ge \frac{E_k}{8k} > 0 \qquad (\delta/2 \le E_k \le 4\delta,\ k \ge K)\,. \tag{82}$$

Globally in (X, V) we have $|J_k - E_k| \le 2eE_k^{3/4}$. Hence, for $k \ge K$,

$$J_k \le \delta \implies E_k < 2\delta, \qquad \delta \le J_k < 2\delta \implies \delta/2 < E_k < 4\delta. \tag{83}$$

For $E \ge 2\delta$, use $E^{1/4} > 1/10$; for $E \le \delta/2$, use $E^{3/4} \le \delta^{3/4} = 1/1000$.

We must also exclude a single jump across this region. Suppose the new state satisfies $E_+ \le 2\delta$. Then $|X_+| < 3/10$ and $|V_+| \le 1/50$. The scalar map $T_k(t) = (1-k^{-2})t - \varphi'(t)/5$ has derivative at least $3/20$ for $k \ge 2$ and fixes zero. Its inverse gives $|Y| \le 2$, where $Y = \sqrt{k}\, y_k$. Thus $|eY| < 1$ and the exact inverse relations are

$$\begin{aligned} X &= X_+/\sqrt{1+e^2} - eV_+/(1+e^2), \\ X_+/\sqrt{1+e^2} &= (1-e^4)Y - e^2Y^3/5, \\ V &= \frac{V_+/(1+e^2) + eY^3/5 + e^3Y}{1 - 5e^2/4}. \end{aligned}$$

For $e \le 1/1000$ they imply

$$|Y| < \frac{301}{1000}, \quad |X - X_+| \le \frac{21}{1000}e, \quad |V - V_+| \le \frac{6}{1000}e, \quad |X| < \frac{301}{1000}, \quad |V| < \frac{21}{1000}.$$

Indeed the second equation first gives $|Y - X_+/\sqrt{1+e^2}| \le 2e^2$; substitution in the third gives the stated velocity bound. On the segment between these states, $|\partial_X E| < 6/1000$ and $|\partial_V E| < 21/1000$. Including the two cross terms defining J gives

$$|J_k - J_{k+1}| < \left[\frac{252}{10^6} + \frac{5}{12}\left(\frac{301 \cdot 21}{10^6} + \frac{3}{500}\right)\right] e < \frac{e}{100} \le \frac{\delta}{10}. \tag{84}$$

This bound controls the preimage of a small new state; no bounded scaled region was presumed for the old state.

It follows that $J_K \ge \delta$ forces $J_k \ge \delta$ for every $k \ge K$. Otherwise, at the first crossing below δ, (83) and (84) give $\delta \le J_k < 2\delta$, putting the old energy in the range of the positive increment (82), a contradiction.

The fixed equal initialization. We verify $J_K > \delta$ for $z_0 = z_1 = R$ without computing the orbit. If $J_K \le \delta$, then $E_K < 2\delta$ and $\max\{|z_{K-1}|, |z_K|\} < 1$. For $j \ge 2$, T_j^{-1} is $20/3$-Lipschitz and $\beta_j \ge 3/8$. Rearranging the actual recurrence gives

$$|z_{j-1}| \le \frac{11}{3}|z_j| + \frac{160}{9}|z_{j+1}|.$$

With $M_j = \max\{|z_{j-1}|, |z_j|\}$ this implies $M_j \le (193/9)M_{j+1} \le 22M_{j+1}$. Iteration from $K-1$ down to 2 would give $|z_1| \le M_2 < 22^{K-2}$, contradicting $z_1 = R = 22^K$. The barrier therefore applies.

Since X_k, V_k are bounded, the correction in (78) tends to zero, so $\liminf E_k \geq \delta > 0$. If $X_k \to 0$, then $|V_k|$ stays away from zero. The exact scaled map gives $\Delta V_k \to 0$, so the sign of V_k is eventually fixed. But $\Delta X_k = V_k/\sqrt{k} + O(k^{-1})$ would then force an unbounded change in X_k, a contradiction. If $V_k \to 0$, the energy floor similarly keeps $|X_k|$ away from zero, and $\Delta X_k \to 0$ fixes its sign. Now $\Delta V_k = -X_k^3/(5\sqrt{k}) + O(k^{-1})$ contradicts boundedness of V_k. Thus neither scaled coordinate tends to zero. On the eventual quartic tail, $k^2\varphi(z_k) = X_k^4/4$, which proves the two positive finite limsups separately. Scaling by the fixed factors R^{-2} and R^{-1} proves (76).

Finally the first component of $\mathcal{F}_R$ has 1/2-Lipschitz derivative and remains zero from its specified initial pair. Its minimizer set and transverse bound follow directly from nonnegativity and $|n| \leq R^{-1}$ on the stated sublevel. The second component is precisely the scalar orbit, proving the remaining assertions. ∎

Both K and R are fixed, not chosen with the observation horizon. The construction makes no claim about a practical transient scale or the sharp damping threshold for all quartic-growth objectives. It shows that the little-o conclusions of Theorem 5(i) cannot include $q = 4$ for every $a > 1$, even with equal unit initial points.

An exact damping threshold for a fixed nonlinear objective

The preceding example separates the two little-o rates from selection. We next classify selection for one coupled objective. The number $3/2 = 1 + 2/4$ is the classical quartic flatness boundary for unregularized inertial rates [4, 10]; the assertion here concerns minimum norm selection for the exact critical recurrence, including equality. The perspective construction is standard [14]. Related oscillation-induced motion occurs in other algorithms [11], and numerical nonselection after removing one regularizer from a different two-term method is reported in [16, Section 5].

Proposition 6.2. Define the single fixed objective

$$\Psi(u,v) = \begin{cases} \dfrac{v^4}{4(2+u)^3}, & -1 \leq u \leq 1, \\ +\infty, & \text{otherwise,} \end{cases} \qquad f(x) = \min_{z\in\mathbb{R}^2} \left\{ \Psi(z) + \frac{1}{2}\|x - z\|^2 \right\}. \tag{85}$$

It is convex and continuously differentiable with globally 1-Lipschitz gradient and bounded sublevels, and

$$S = [-1,1] \times \{0\}, \qquad A = \operatorname{aff} S = \mathbb{R} \times \{0\}, \qquad x^* = 0.$$

On $f \leq 1$, $f(x) \geq \operatorname{dist}(x, A)^4/896$. No positive transverse lower bound of order less than four holds on any positive low sublevel.

Keep $b = 1$ and $s = 1/5$ in (1). For every fixed $a > 0$ and every initial pair, both orbits are bounded and their distances to S tend to zero. For $a > 1$ they also have objective residual $O(k^{-2})$ and velocity $O(k^{-1})$. Their selection properties are as follows:

(i) If $a \geq 3/2$, every initial pair satisfies $x_k, y_k \to 0$, $f(x_k) = o(k^{-2})$, and $\|x_k - x_{k-1}\| = o(k^{-1})$.

(ii) If $1 < a < 3/2$, the pair $x_0 = x_1 = (0, 1)$ satisfies $\limsup_k \|x_k\| > 0$.

(iii) If $0 < a \leq 1$, the pair $x_0 = x_1 = (1, 0)$ satisfies $\limsup_k \|x_k\| > 0$.

Thus universal minimum norm selection for this fixed f, b, s holds exactly when $a \geq 3/2$. The targeted defect (6) is identically zero.

Proof. *Global geometry, including boundary contact.* The finite branch of Ψ is the perspective of $v \mapsto v^4/4$, restricted to a closed strip with positive denominator. Thus Ψ is proper, lower semicontinuous and convex. Its proximity operator $P = \mathrm{prox}_\Psi$ is single-valued and firmly nonexpansive. The identity $\nabla f = I - P$ and firm nonexpansiveness of $I - P$ give the global Lipschitz bound. Nonnegativity of the two terms in (85) gives the stated S.

At $x = (t, n)$ write $P(x) = (u, v)$, $W = 2 + u \in [1, 3]$, $r = v/W$, and $g = \nabla f(x) = (g_t, g_n)$. The exact proximal relations are

$$\begin{gathered} g_n = r^3, \qquad g_t = -\frac{3}{4}r^4 + \xi, \quad \xi \in N_{[-1,1]}(u), \\ t = u + g_t, \qquad n = Wr + r^3, \qquad f(x) = \frac{W}{4}r^4 + \frac{1}{2}(g_t^2 + r^6). \end{gathered} \tag{86}$$

Since $u\xi \geq 0$ and $(W - 3u)r^4/4 = (1 - u)r^4/2 \geq 0$, we have

$$tg_t \geq -f(x) + \frac{3}{2}g_t^2 + \frac{1}{2}g_n^2, \qquad ng_n = 4f(x) - 2g_t^2 - g_n^2. \tag{87}$$

These hold even when the proximal point is at a strip endpoint.

For $H = \mathrm{diag}(h_t, h_n) > 0$, put

$$\kappa_0 = 4h_n - h_t, \qquad A_0 = \max\{0, h_n - h_t/2, 2h_n - 3h_t/2\}.$$

Then

$$\langle Hx, \nabla f(x)\rangle \geq \kappa_0 f(x) - A_0 \|\nabla f(x)\|^2 . \tag{88}$$

A second global estimate follows from $W \leq 3$ and $g_n = r^3$:

$$f(x) \leq \frac{3}{4}\|\nabla f(x)\|^{4/3} + \frac{1}{2}\|\nabla f(x)\|^2 . \tag{89}$$

If $f(t, n) \leq M$, then $|u| \leq 1$, $|v|^4 \leq 108M$ and $(t - u)^2 + (n - v)^2 \leq 2M$. Hence sublevels are bounded. On $f \leq 1$,

$$|n|^4 \leq 8|v|^4 + 8|n - v|^4 \leq 864f + 32f^2 \leq 896f.$$

The test point $(u, v) = (0, n)$ gives $f(0, n) \leq n^4/32$, proving failure of every smaller transverse exponent. Lemma 2 and finite-dimensional compactness now prove boundedness

and distance to S tending to zero for every $a > 0$. All finite initial steps exist because the gradient is globally defined.

An anisotropic energy estimate. Fix $a > 1$, $\mu = a - 1$, a positive diagonal matrix H, and $D = \mu I - H > 0$. Suppose that fixed constants $\kappa > 2$ and $A, B \geq 0$ satisfy

$$\langle Hx, \nabla f(x)\rangle \geq \kappa f(x) - A\,\|\nabla f(x)\|^2 - B\,\|\nabla f(x)\|^{4/3} \qquad (x \in \mathbb{R}^2). \tag{90}$$

Let h_-, h_+ and d_-, d_+ be the extreme eigenvalues of H and D. Define

$$\begin{aligned} &G_k = \nabla f(y_k) + \frac{y_k}{sk^2}, \qquad U_k = Hx_k + (k-a)v_k, \qquad \omega_k = \left(\frac{k-1}{k}\right)^2, \\ &\mathcal{E}_k = s(k-1)(k-1+\kappa)f(x_k) + \frac{\omega_k}{2}\,\|x_k\|^2 + \frac{1}{2}\,\|U_k\|^2 + \frac{1}{2}\,\langle x_k, HDx_k\rangle\,. \end{aligned} \tag{91}$$

Every term is nonnegative for $k \geq 2$. Set $\rho = 1 - s = 4/5$, $\sigma = 1 - s/2 = 9/10$, and $B_y = \sup_k \|y_k\| < \infty$.

For one step abbreviate $x = x_k$, $p = v_{k+1}$, $y = y_k$ and $G = G_k$. Since $U_k = Hx + kp + skG$ and $U_{k+1} = Hx + (kI - D)p$, the exact kinetic identity is

$$\Delta\left(\frac{\|U\|^2}{2} + \frac{\langle x, HDx\rangle}{2}\right) = -sk\,\langle Hx + kp, G\rangle - \frac{s^2k^2}{2}\,\|G\|^2 - \left(k - \frac{\mu}{2}\right)\langle p, Dp\rangle\,. \tag{92}$$

Apply smoothness and strong convexity to $\Phi_k = f + \|\cdot\|^2/(2sk^2)$, whose gradient at y is G. If $\mathcal{E}^0$ denotes (91) without the linear objective correction $s\kappa(k-1)f(x_k)$, the result is

$$\begin{aligned} \Delta\mathcal{E}^0 \leq{}& s(2k-1)f(x) - sk\,\langle Hx, G\rangle - \frac{s^2k^2}{2}\left(\rho - \frac{1}{k^2}\right)\|G\|^2 \\ &- \left(k - \frac{\mu}{2}\right)\langle p, Dp\rangle - \frac{1}{2}\,\|\beta_k v_k\|^2 + R_{\mathrm{pos}}, \\ R_{\mathrm{pos}} ={}& \frac{2k-1}{2k^2}\,\|x\|^2 - \frac{2k+1}{2(k+1)^2}\,\|x+p\|^2\,. \end{aligned} \tag{93}$$

All changes of the regularization coefficient are included here. Moreover, $\beta_k v_k = p + sG$ gives exactly

$$-sk\,\langle Hx, G\rangle = -sk\,\langle Hy, \nabla f(y)\rangle + sk\,\langle Hp, G\rangle + s^2k\,\langle HG, G\rangle - \frac{1}{k}\,\langle Hy, y\rangle\,.$$

Using (90) and restoring the linear correction makes the objective contribution

$$-s[(\kappa-2)k - (\kappa-1)]f(x) + s\kappa k[f(x+p) - f(y)].$$

This explains the weight in (91); no scalar flatness comparison is presumed for H.

The smooth step and completing the square give

$$f(x+p) - f(y) \leq \frac{B_y^2}{4s\sigma k^4}, \qquad sk|\,\langle Hp, G\rangle\,| \leq \frac{d_-}{8}k\,\|p\|^2 + \frac{2s^2h_+^2}{d_-}k\,\|G\|^2\,.$$

Writing $R_{\mathrm{pos}} = r_k \|x\|^2 - 2c_k \langle x, p\rangle - c_k \|p\|^2$, we have $0 \le r_k \le k^{-2}$ and $0 \le c_k \le k^{-1}$. For $k \ge 8/h_-$, Young's inequality and $y = x + p + sG$ yield

$$R_{\mathrm{pos}} - \frac{1}{k}\langle Hy, y\rangle \le -\frac{h_-}{4k}\|x\|^2 + \frac{2h_- + 8/h_-}{k}\|p\|^2 + \frac{2h_- s^2}{k}\|G\|^2 .$$

Also $sAk\|\nabla f(y)\|^2 \le 2sAk\|G\|^2 + 2AB_y^2/(sk^3)$. These terms can be absorbed on a finite tail: it suffices to impose $k \ge 2\mu$, $k^2 \ge 2/\rho$,

$$k^2 \ge \frac{8(2h_- + 8/h_-)}{d_-}, \qquad \left(2sA + \frac{2s^2h_+^2}{d_-} + s^2h_+\right)k + \frac{2h_-s^2}{k} \le \frac{\rho s^2}{8}k^2,$$

and $k \ge 2(\kappa-1)/(\kappa-2)$. Every cutoff is finite for the fixed parameters.

Put $Q = 2^{1/3}sB$ and $\epsilon_g = \rho s^2/16 = 1/500$. The remaining defect obeys

$$sBk\|\nabla f(y)\|^{4/3} \le Qk\|G\|^{4/3} + Q(B_y/s)^{4/3}k^{-5/3},$$

and exact scalar maximization gives

$$Qku^{4/3} - \epsilon_g k^2 u^2 \le \frac{C_p}{k}, \qquad C_p = \frac{4Q^3}{27\epsilon_g^2}, \qquad u \ge 0.$$

Consequently

$$\begin{aligned} \Delta\mathcal{E}_k \le\ & -\frac{s(\kappa-2)}{2}kf(x_k) - \frac{h_-}{4k}\|x_k\|^2 - \frac{d_-}{2}k\|v_{k+1}\|^2 - \frac{\rho s^2}{16}k^2\|G_k\|^2 + R_k, \\ R_k =\ & \frac{C_0}{k^3} + Q(B_y/s)^{4/3}k^{-5/3} + \frac{C_p}{k}, \qquad C_0 = \frac{\kappa B_y^2}{4\sigma} + \frac{2AB_y^2}{s}. \end{aligned} \tag{94}$$

The exact U_k identity bounds the energy by

$$\mathcal{E}_k \le s(1+\kappa)k^2 f(x_k) + C_x\|x_k\|^2 + 2k^2\|v_{k+1}\|^2 + 2s^2k^2\|G_k\|^2, \qquad C_x = h_+^2 + \frac{1}{2} + \frac{h_+d_+}{2}.$$

Thus, for

$$\theta = \min\left\{1, \frac{\kappa-2}{2(1+\kappa)}, \frac{h_-}{4C_x}, \frac{d_-}{4}, \frac{\rho}{32}\right\} > 0, \qquad \mathcal{E}_{k+1} \le \left(1 - \frac{\theta}{k}\right)\mathcal{E}_k + R_k. \tag{95}$$

The harmonic comparison proves boundedness and $\limsup_k \mathcal{E}_k \le C_p/\theta$, because $k(R_k - C_p/k) \to 0$. No uniform starting index in auxiliary parameters is needed.

Upper rates and the open positive range. For an arbitrary fixed $a > 1$, put $h = (a-1)/2$, $H = D = hI$, and $c = (3-3h)_+$. Equations (88)–(89) give (90) with

$$\kappa = 3, \qquad A = h/2 + c/2, \qquad B = 3c/4.$$

Boundedness of (91) proves both O rates, before selection is considered.

If $a > 3/2$, set $\xi = (a - 3/2)/4 > 0$ and choose

$$H = \mathrm{diag}(\xi, 1/2+\xi), \qquad D = \mathrm{diag}(1/2+3\xi, 3\xi), \qquad \kappa = 2+3\xi, \quad A = 1+\xi/2, \quad B = 0.$$

Now (88) supplies the comparison directly. Since $C_p = 0$, (95) gives $\mathcal{E}_k \to 0$. The position and objective terms yield $x_k \to 0$ and $k^2 f(x_k) \to 0$. From $(k-a)v_k = U_k - Hx_k$ we obtain $kv_k \to 0$, and hence $y_k \to 0$.

The equality case $a = 3/2$. Keep the actual recurrence fixed. For an auxiliary $0 < \varepsilon \le 1/8$, let

$$H_\varepsilon = \operatorname{diag}(\varepsilon, 1/2 - \varepsilon), \qquad D_\varepsilon = \operatorname{diag}(1/2 - \varepsilon, \varepsilon).$$

Equation (88) has $\kappa_0 = 2 - 5\varepsilon$ and $A_0 = 1 - 7\varepsilon/2$. Apply (89) to increase its objective coefficient by 6ε. This gives the global comparison

$$\langle H_\varepsilon x, \nabla f(x)\rangle \ge (2+\varepsilon) f(x) - \|\nabla f(x)\|^2 - \frac{9\varepsilon}{2}\|\nabla f(x)\|^{4/3}.$$

The quadratic coefficient actually needed is $1 - \varepsilon/2 \le 1$. Thus the preceding energy estimate applies with $\kappa = 2 + \varepsilon$, $A = 1$ and $B = 9\varepsilon/2$. Here $h_- = d_- = \varepsilon$, $h_+, d_+ \le 1/2$ and $C_x \le 7/8$. In (95), each entry is at least $\varepsilon/8$. The exact coefficients therefore satisfy

$$\theta_\varepsilon \ge \frac{\varepsilon}{8}, \qquad C_p = 54000\varepsilon^3, \qquad \limsup_k \mathcal{E}_k^{(\varepsilon)} \le 432000\varepsilon^2. \tag{96}$$

The starting index may diverge as $\varepsilon \downarrow 0$; the time limit in this display is taken first.

The position coefficient $\omega_k/2$ tends to $1/2$ independently of ε, and the objective coefficient divided by k^2 tends to s. Hence (96), followed by $\varepsilon \downarrow 0$, proves $x_k \to 0$ and $k^2 f(x_k) \to 0$. The kinetic term and $\|H_\varepsilon\| \le 1/2$ give $kv_k \to 0$ by the same ordered limits. Extrapolation gives $y_k \to 0$. This proves the equality case without changing any algorithm parameter or assuming a uniform transient energy bound.

Nonselection for $1 < a < 3/2$. Fix such an a and the pair $x_0 = x_1 = (0, 1)$. The upper rates just proved give

$$w_k = k(t_k - t_{k-1}) = O(1), \qquad p_k = k(n_k - n_{k-1}) = O(1), \qquad n_k = O(k^{-1/2}). \tag{97}$$

Local formulas under hypothetical selection. Suppose, for contradiction, that $x_k \to 0$. Then $y_k \to 0$. Near zero the proximal point is in the interior of the strip, since P is continuous and $P(0) = 0$. Put $P(t, n) = (u, v)$, $W = 2 + u$, and $r = v/W$. The exact optimality equations are

$$t = u - \frac{3}{4}r^4, \qquad n = Wr + r^3, \qquad W = 2 + t + \frac{3}{4}r^4, \qquad f_t = -\frac{3}{4}r^4, \quad f_n = r^3. \tag{98}$$

The equation $n = (2 + t + 3r^4/4)r + r^3$ defines an analytic local function. On $|t| \le 1/2$, $|n| \le 3/8$, its solution has $|r| \le 1/4$, and $u = t + 3r^4/4$ lies strictly inside the strip. Put

$$D_r = 2 + t + 3r^2 + \frac{15}{4}r^4 \ge \frac{3}{2}.$$

Exact differentiation and the envelope value give

$$f(t, n) = \frac{W}{4}r^4 + \frac{1}{2}r^6 + \frac{9}{32}r^8, \qquad f_{nn} = \frac{3r^2}{D_r}, \quad f_{nt} = -\frac{3r^3}{D_r}, \quad f_{tt} = \frac{3r^4}{D_r},$$

$$n f_n - 4f = -r^6 - \frac{9}{8}r^8.$$

Since $|n|/3 \le |r| \le 2|n|/3$ there, fixed positive constants c, C satisfy

$$\begin{aligned} cn^4 \le f(t,n) \le Cn^4, \qquad |f_n| \le C|n|^3, \quad |f_t| \le C|n|^4, \\ |f_{nn}| \le Cn^2, \quad |f_{nt}| \le C|n|^3, \quad |f_{tt}| \le Cn^4, \qquad |nf_n - 4f| \le C|n|^6. \end{aligned} \tag{99}$$

One may take $c = 1/216$ and $C = 1$ in these bounds on the stated rectangle. In particular $f = n^4/[4(2+t)^3] + O(n^6)$ and $f_n = n^3/(2+t)^3 + O(n^5)$ uniformly there.

Let r_k be the proximal parameter at y_k. The tangential recurrence is

$$w_{k+1} - w_k = \frac{(1-a)w_k - t_k}{k} + \frac{3}{20} k r_k^4 + O(k^{-2}). \tag{100}$$

Since $r_k = O(k^{-1/2})$, we obtain $|w_{k+1} - w_k| \le C/k$. This and $t_k \to 0$ imply $w_k \to 0$. Indeed, any late value $|w_m| \ge \epsilon$ persists with the same sign and magnitude at least $\epsilon/2$ on a fixed short multiplicative interval $[m, (1+\theta)m]$. Summing $t_{j+1} - t_j = w_{j+1}/(j+1)$ there contradicts convergence of t_k. Summing (100) on $[m, 2m]$ now gives

$$\sum_{k=m}^{2m} k r_k^4 \longrightarrow 0. \tag{101}$$

Define the normal potential and energy at the actual point by

$$V_k = sk^2 f(t_k, n_k), \qquad E_k = \frac{1}{2} p_k^2 + V_k, \qquad B_k = n_k p_k. \tag{102}$$

They retain the changing tangential coordinate. The local formulas and the O rates show that $k^2[f(y_k) - f(x_k)] \to 0$ and $k^2 r_k^4 = 10 V_k + o(1)$. Consequently

$$\sum_{k=m}^{2m} \frac{V_k}{k} \longrightarrow 0. \tag{103}$$

Uniform actual-step estimates. For the old state suppress k and write $D_0 = p^2 + n^2$. Direct expansion of the recurrence gives

$$\begin{aligned} \Delta E &= \frac{2V + (1-a)p^2 - np}{k} + \mathcal{R}_E, \\ \Delta B &= \frac{p^2 + (1-a)np - n^2 - 4V}{k} + \mathcal{R}_B, \\ \Delta(n^2/2) &= \frac{np}{k} + \mathcal{R}_N, \end{aligned} \tag{104}$$

where a fixed C on the contradiction tail satisfies

$$|\mathcal{R}_E| + |\mathcal{R}_B| + |\mathcal{R}_N| \le \frac{C}{k} \left[\left(|w| + \sqrt{E} + \frac{1}{k} \right) E + \frac{D_0}{k} \right]. \tag{105}$$

No lower bound on E is used here. We give the details needed for that uniformity.

Put $N = |n| + |p|/k$. Boundedness of E and (99) imply $N^2 \le C\sqrt{E}/k$ and $p^2 \le 2E$. If $\alpha = t_{k+1} - t_k$ and $\nu = n_{k+1} - n_k$, then

$$\begin{aligned} \nu &= p/k + O((|p| + |n|)/k^2 + N^3), \\ \alpha &= w/k + O((|w| + |t|)/k^2 + N^4), \\ p_{k+1} - p &= ((1-a)p - n)/k - skf_n(t,n) + r_p, \\ |r_p| &\le C[(|p| + |n|)/k^2 + N^3 + N^2|p| + N^3|w|]. \end{aligned}$$

The bound on r_p follows by comparing $f_n(y_k)$ with $f_n(x_k)$ using the second derivatives in (99). In $\Delta(p^2/2)$ the term $-skpf_n$ cancels its opposite in $\Delta(sk^2 f)$. The changing weight contributes $2V/k$, and the tangential term has magnitude at most $C|w|V/k$. The remaining terms obey

$$\begin{gathered} p^2N^2 \le CE^{3/2}/k, \qquad |p|N^3 \le CE^{5/4}/k^{3/2} \le (C/k)(\sqrt{E} + 1/k)E, \\ k^2N^6 \le CE^{3/2}/k, \qquad N^4 \le CE/k^2, \qquad k^2N^8 \le CE^2/k^2 \le (C/k)(\sqrt{E} + 1/k)E. \end{gathered}$$

The second-order potential remainder is bounded by $Ck^2[N^2\nu^2 + N^3|\alpha\nu| + N^4\alpha^2]$. The remaining quadratic products contribute at most CD_0/k^2; the products $(|p| + |n|)N^3$ are covered by the preceding bounds. These estimates prove the first line of (104) with (105). Expanding $\Delta B = n\Delta p + p\nu + \nu\Delta p$ and using $nf_n = 4f + O(n^6)$ proves the second line; the identity $\Delta(n^2/2) = n\nu + \nu^2/2$ proves the third. The sharper absolute bounds $n, N = O(k^{-1/2})$, $\nu = O(k^{-1})$, and $\Delta p = O(k^{-1/2})$ also give

$$\mathcal{R}_B = O(k^{-3/2}), \qquad \mathcal{R}_N = O(k^{-2}).$$

In particular, the error estimate vanishes with the normal state; it has no fixed additive remainder that could dominate an arbitrarily small energy.

Since $B_k = O(k^{-1/2})$, the second line gives the virial relation

$$\Delta B_k = \frac{p_k^2 - 4V_k}{k} + O(k^{-3/2}).$$

Summation on $[m, 2m]$, followed by (103), yields

$$\sum_{k=m}^{2m} \frac{E_k}{k} \longrightarrow 0. \tag{106}$$

A first correction forces vanishing energy. Set $J_k = E_k + (a/3)B_k$ and $\lambda = 2 - 4a/3 > 0$. Equations (104) give

$$\Delta J_k = \frac{1}{k}\left[\lambda E_k + \left(\frac{a(1-a)}{3} - 1\right)B_k - \frac{a}{3}n_k^2\right] + \mathcal{R}_J. \tag{107}$$

where $\mathcal{R}_J$ satisfies (105) after enlarging C. We have $J_k - E_k \to 0$ and $|\Delta J_k| = O(k^{-1})$.

For every sufficiently small fixed $\delta > 0$, a sufficiently late tail satisfies

$$\delta \le J_k \le 2\delta \quad \Longrightarrow \quad \delta/2 \le E_k \le 3\delta, \qquad \Delta J_k > 0.$$

Indeed, on this fixed energy interval,

$$\frac{|B_k|}{E_k} \le Ck^{-1/2}E_k^{-1/4} \to 0, \qquad \frac{n_k^2}{E_k} \le \frac{C}{k\sqrt{E_k}} \to 0.$$

First choose δ so that the $C\sqrt{E}$ error is small; then use $w_k \to 0$ and a late enough index for the other terms in (107). This gives, for example, $\Delta J_k \ge \lambda E_k/(2k)$.

If $\limsup J_k > 0$, choose such a δ with $\limsup J_k > 2\delta$. Equation (106) supplies arbitrarily late values below δ. A first subsequent crossing below δ from above 2δ has its old value in $[\delta, 2\delta]$, because $\Delta J_k \to 0$. This contradicts the positive increment there. Hence

$$E_k \longrightarrow 0. \tag{108}$$

A second energy excludes an exceptional small tail. Put

$$Q_k = \frac{p_k^2 + n_k^2}{2} + \frac{a-1}{2}B_k, \qquad P_k = Q_k + V_k.$$

Then

$$\frac{3}{8}(p_k^2 + n_k^2) \le Q_k \le \frac{5}{8}(p_k^2 + n_k^2), \qquad E_k \le \frac{4}{3}P_k.$$

Using all three lines of (104) gives

$$\Delta P_k = \frac{-(a-1)Q_k + (4-2a)V_k}{k} + \mathcal{R}_P.$$

By (105), $w_k \to 0$, and (108), the error is $o(P_k/k)$. Since $a - 1 < 1/2$ and $4 - 2a > 0$, we obtain

$$P_{k+1} \ge \left(1 - \frac{1}{k}\right) P_k \tag{109}$$

on a tail.

The starting energy on that tail is strictly positive for the actual initial pair. The function f is even in n, so $f_n(t, 0) = 0$. For fixed t and $k \ge 2$, convexity and the global Lipschitz bound imply that

$$n \longmapsto (1 - k^{-2})n - \tfrac{1}{5}f_n(t, n)$$

is strongly increasing with constant at least $11/20$ and has its unique zero at zero. If $n_k = n_{k+1} = 0$, the update forces $(y_k)_n = 0$, and $1 - a/k > 0$ forces $n_{k-1} = 0$. Repetition contradicts $n_1 = 1$. Thus no adjacent normal pair vanishes and $P_k > 0$ on every late tail. The product in (109) consequently gives

$$P_k \ge c_N k^{-1}, \qquad c_N > 0,$$

where c_N depends on the actual positive tail value. On the other hand,

$$n_k^2 \le C\sqrt{E_k}/k, \qquad P_k \le C(E_k + n_k^2) \le C'(E_k + k^{-2}).$$

It follows that $E_k \ge ck^{-1}$ eventually. In particular,

$$\frac{|B_k|}{E_k} = O(k^{-1/4}) \to 0, \qquad \frac{n_k^2}{E_k} = O(k^{-1/2}) \to 0.$$

Now $J_k/E_k \to 1$, and every error in (107), divided by E_k/k, tends to zero. Thus $J_k > 0$ and $\Delta J_k \ge \lambda E_k/(2k) > 0$ eventually. This contradicts $J_k \to 0$ from (108).

Selection to zero is therefore impossible in this open range. Boundedness gives the stated positive limsup.

The lower range $0 < a \le 1$. Take $x_0 = x_1 = (1, 0)$. The normal coordinate remains zero, and

$$f(t, 0) = \frac{1}{2}\operatorname{dist}(t, [-1, 1])^2, \qquad f_t(t, 0) = t - P_{[-1,1]}t.$$

At $a = 1$, the auxiliary flat orbit in Proposition 5.1 has every extrapolated point in $[-1, 1]$. Its gradient is therefore zero for the present objective as well, so it is exactly this tangent orbit, with the same $b = 1$. The invariant bound $41/96 \le J_k \le 41/72$ and the ensuing harmonic-sum contradiction prove that it does not tend to zero.

For $0 < a < 1$, the scalar map

$$t \longmapsto (1 - k^{-2})t - \tfrac{1}{5}(t - P_{[-1,1]}t)$$

is strongly increasing with constant at least $11/20$ for $k \ge 2$, and has unique zero zero. Since $\beta_k > 0$, an adjacent zero pair would propagate backwards to contradict $t_1 = 1$. If the actual tangent orbit converged to zero, its entire late extrapolated tail would lie in the interval. That tail would be a nonzero solution of the flat recurrence (128). The positive matrix identity (129)–(130), followed by the sign and harmonic-sum argument (131), excludes such convergence for $0 < a < 1$. This proves the last range.

Finally, $p = -x^* = 0$, $f^*(0) = 0$ and $\sigma_S(0) = 0$, so the targeted defect vanishes. The positive and negative ranges establish the stated equivalence for the single fixed objective and the fixed b, s. ∎

The universal quantifier concerns initial data: the lower ranges supply the specified failing pairs, while the positive range covers every pair. No threshold for other objectives, regularization coefficients or stepsizes is asserted.

A continuous endpoint with different selection behavior

High-resolution analyses identify additional gradient and Hessian-driven damping in accelerated algorithms [7, 19]. For the fixed objective above, the following comparison concerns the selected point itself. We use iteration-index time in the continuous equation; a constant rescaling gives the usual unit-gradient normalization.

Proposition 6.3. Let f be the objective in (85), and consider

$$\ddot{x}(t) + \frac{3}{2t}\dot{x}(t) + \frac{1}{5}\nabla f(x(t)) + \frac{1}{t^2}x(t) = 0, \qquad x(1) = (0,1), \quad \dot{x}(1) = 0. \tag{110}$$

Its unique solution is defined for every $t \geq 1$, both x and $\dot{x}$ are bounded, and

$$0 < \limsup_{t\to\infty} \|x(t)\| < \infty.$$

Thus this continuous trajectory fails to select the minimum norm point, whereas the recurrence of Proposition 6.2 with $a = 3/2$, $b = 1$, $s = 1/5$ selects that point from every initial pair, including $x_0 = x_1 = (0,1)$.

Proof. Put $s = 1/5$. The gradient is globally Lipschitz. The mechanical energy

$$M(t) = sf(x(t)) + \frac{1}{2}\|\dot{x}(t)\|^2 + \frac{1}{2t^2}\|x(t)\|^2$$

satisfies $\dot{M} = -3\|\dot{x}\|^2/(2t) - \|x\|^2/t^3 \leq 0$. Bounded sublevels of f give boundedness of x and $\dot{x}$, hence global existence. Since f is even in its second coordinate, the normal position and velocity cannot both vanish at a finite time: uniqueness backwards would give a zero normal coordinate at $t = 1$.

Suppose, for contradiction, that $x(t) \to 0$. Use logarithmic time $\sigma = \log t$ and write $(T, N) = x(e^\sigma)$, $w = T'$ and $p = N'$. All following primes denote derivatives in σ. The proximal point is eventually interior in the strip. With $W = 2 + T$ and the local variable r in (98), the exact identities are

$$\begin{aligned} &N = Wr + r^3 + \frac{3}{4}r^5, \qquad && f_N = r^3, \qquad && f_T = -\frac{3}{4}r^4, \\ &f = \frac{W}{4}r^4 + \frac{1}{2}r^6 + \frac{15}{32}r^8, \quad && Nf_N - 4f = -r^6 - \frac{9}{8}r^8. \end{aligned} \tag{111}$$

On a fixed small rectangle, W is bounded above and away from zero, $|r|$ is comparable to $|N|$, and $cN^4 \leq f \leq CN^4$.

The tangential equation has one-sided forcing:

$$w' + \frac{1}{2}w = -T + F, \qquad F = \frac{3s}{4}e^{2\sigma}r^4 \geq 0. \tag{112}$$

It follows directly that $w \to 0$. Indeed, on a tail with $|T| \leq \epsilon$, scalar comparison gives $\liminf w \geq -2\epsilon$. If $w(\sigma_j) \geq \delta > 0$, the same comparison keeps $w \geq \delta/2$ on a fixed short forward interval, once ϵ is small enough. This forces a fixed positive change in T, contrary to $T \to 0$. Integration of (112) therefore yields

$$\int_\sigma^{\sigma+1} F(u)\,du \longrightarrow 0. \tag{113}$$

No bound on a scaled normal velocity has been used.

Define the physical normal quantities

$$V = se^{2\sigma} f(T,N), \quad E = \frac{1}{2}p^2 + V, \quad B = Np,$$

$$H = se^{2\sigma}\left(r^6 + \frac{9}{8}r^8\right), \qquad K = \frac{3se^{2\sigma}}{W}\left(\frac{1}{2}r^6 + \frac{15}{32}r^8\right).$$

Direct differentiation of the actual equation gives

$$\begin{aligned} E' &= 2V - \frac{1}{2}p^2 - B - \frac{3w}{W}V + wK, \\ B' &= p^2 - 4V - \frac{1}{2}B - N^2 + H, \qquad F = \frac{3V}{W} - K. \end{aligned} \tag{114}$$

The local bounds imply, uniformly including zero energy,

$$N^2 \le Ce^{-\sigma}\sqrt{E}, \qquad |B| \le Ce^{-\sigma/2}E^{3/4}, \qquad 0 \le H + K \le CN^2V \le Ce^{-\sigma}E^{3/2}. \tag{115}$$

We first establish a mild upper bound. For $J = E + B/2$, equations (114) give

$$J' = -\frac{5}{4}B - \frac{1}{2}N^2 - \frac{3w}{W}V + wK + \frac{1}{2}H.$$

Since $N, w \to 0$, $H + K \le CN^2E$, and $|B| \le \sqrt{2}|N|\sqrt{E}$, for every $\eta > 0$ a sufficiently late tail has $J' \le \eta(E+1)$. Moreover, $|B|/2 \le E/2 + N^2/4$, so $J + 1 \ge (E+1)/2$ eventually. Gronwall's inequality, with an arbitrarily small η, gives

$$E(\sigma) \le C_\delta e^{\delta\sigma} \quad \text{for every fixed } \delta > 0. \tag{116}$$

Now put $c = (W - w)/2$ and $A = WE + cB$. Since $c' = 3w/4 + T/2 - F/2$, all leading potential and kinetic terms cancel exactly:

$$A' = (-W - c/2 + c')B - cN^2 + cH + WwK. \tag{117}$$

Here T, w, W are bounded and $F \le CE$. Hence

$$|A'| \le Ce^{-\sigma/2}(E^{3/4} + E^{7/4}) + Ce^{-\sigma}(E^{1/2} + E^{3/2}).$$

Taking $\delta = 1/4$ in (116) makes this bound integrable; its slowest exponential factor is $e^{-\sigma/16}$. Also $B \to 0$ and $W \to 2$. Thus A has a finite limit and E has a finite nonnegative limit. Now $H, K \to 0$. Equations (113) and (114) first give $\int_\sigma^{\sigma+1} V \to 0$, and then, by integrating B', give $\int_\sigma^{\sigma+1} p^2 \to 0$. Consequently

$$E \longrightarrow 0. \tag{118}$$

The actual nonzero normal state supplies the complementary lower bound. Let

$$Q = \frac{1}{2}(p^2 + N^2) + \frac{1}{4}B, \qquad P = Q + V.$$

The form Q is positive definite, with $3(p^2+N^2)/8 \le Q \le 5(p^2+N^2)/8$. The exact derivative is

$$P' = -\frac{1}{2}Q + V - \frac{3w}{W}V + wK + \frac{1}{4}H.$$

The last three terms are $o(P)$, so eventually $P' \ge -3P/4$. Its initial value on that tail is strictly positive by uniqueness of the actual normal state. Therefore $P \ge ce^{-3\sigma/4}$. On the other hand, (115) and Young's inequality give $P \le CE + C'e^{-2\sigma}$. It follows that

$$E \ge c'e^{-3\sigma/4}, \qquad \frac{|B|}{E} \le Ce^{-5\sigma/16}, \qquad \frac{N^2}{E} \le Ce^{-5\sigma/8}$$

on a later tail. Since E is now bounded, (117) yields $|A'| \le R(\sigma)E$ with $\int^\infty R < \infty$. Furthermore $A/(WE) \to 1$, so $A > 0$ and $E \le CA$ eventually. Integrating $|A'| \le CRA$ gives a strictly positive lower bound on A, contradicting (118) and $B \to 0$. This proves nonselection, and global boundedness gives the asserted positive finite limsup. ∎

This comparison specifies the function, both evolutions and both initial states. It asserts neither a particular nonzero continuous limit nor a rate for that trajectory. The conclusion does not require a uniform approximation of the discrete orbit by the continuous one on an unbounded time interval.

7 Proof of Theorem 6

The minimizer set of (10) is $x^* + \ker Q$, so $x^* \perp \ker Q$ identifies its minimum norm point. The exact update is

$$x_{k+1} = \left(I - sQ - \frac{b}{k^2}I\right)\left[x_k + \left(1 - \frac{a}{k}\right)(x_k - x_{k-1})\right] + sQx^*. \tag{119}$$

We first obtain a scalar bound that is uniform down to spectral value zero.

The moving-center scalar energy

For $\lambda \in [0, L]$ and $\xi \in \mathbb{R}$, write the scalar recurrence as

$$u_{k+1} = q_k(u_k + \beta_k v_k) + s\lambda\xi, \quad v_k = u_k - u_{k-1}, \tag{120}$$

where, within this section,

$$t_k = s\lambda + \frac{b}{k^2}, \qquad q_k = 1 - t_k, \qquad \beta_k = 1 - \frac{a}{k}, \qquad z_k = \frac{s\lambda}{t_k}\xi. \tag{121}$$

These local scalar symbols are distinct from the energy notation in Section 2. Since $t_k z_k = s\lambda\xi$,

$$u_{k+1} - z_k = q_k(u_k - z_k + \beta_k v_k), \qquad v_{k+1} = -t_k(u_k - z_k) + q_k\beta_k v_k.$$

Expanding the squares cancels their cross terms and gives the exact identity

$$(u_{k+1} - z_k)^2 + \frac{q_k}{t_k} v_{k+1}^2 = q_k (u_k - z_k)^2 + \frac{q_k^2 \beta_k^2}{t_k} v_k^2. \tag{122}$$

Suppose for now that $a \geq 1$. Set $\rho = 1 - sL > 0$ and choose an integer

$$K \geq \max\left\{2, 2a, 1 + \sqrt{\frac{2b}{\rho}}, 1 + \frac{6}{\rho}\right\}. \tag{123}$$

For every $k \geq K$ and every $\lambda \in [0, L]$, we have $q_k \geq q_{k-1} \geq \rho/2 > 0$. Also

$$t_{k-1} - t_k \leq t_k \frac{2k-1}{(k-1)^2} \leq t_k \frac{3}{k-1} \leq t_k \frac{\rho}{2} \leq t_k q_k.$$

Consequently $q_k^2 \leq q_{k-1}$. A second uniform comparison is

$$\beta_k^2 \frac{t_{k-1}}{t_k} \leq \left(\frac{k-a}{k-1}\right)^2 \leq 1. \tag{124}$$

Here $t_{k-1}/t_k \leq k^2/(k-1)^2$ follows from $\lambda \geq 0$, and $k \geq 2a$ makes the final comparison valid.

Define

$$\gamma_k := \frac{q_{k-1}}{t_{k-1}}, \qquad J_k := (u_k - z_{k-1})^2 + \gamma_k v_k^2.$$

Equations (122) and (124) imply

$$J_{k+1} \leq (u_k - z_k)^2 + \gamma_k v_k^2.$$

The triangle inequality for the weighted Euclidean norm therefore gives

$$\sqrt{J_{k+1}} \leq \sqrt{J_k} + |z_k - z_{k-1}|. \tag{125}$$

For fixed $\lambda > 0$, z_k moves monotonically along the segment from 0 to ξ; for $\lambda = 0$ it is identically zero. Thus

$$\sum_{k=K}^{\infty} |z_k - z_{k-1}| \leq |\xi|, \qquad \sup_{k \geq K} \sqrt{J_k} \leq \sqrt{J_K} + |\xi|.$$

Since $\gamma_K \leq (K-1)^2/b$, it follows that

$$\sup_{k \geq K} |u_k| \leq |u_K| + \frac{K-1}{\sqrt{b}} |v_K| + 3|\xi|. \tag{126}$$

This uses variation of each scalar center. It does not assume bounded variation in norm of a Hilbert-space regularization path.

There is consequently a constant $C = C(a, b, s, L)$ such that

$$\sup_{\lambda \in [0,L],\ k \geq 0} |u_k| \leq C(|u_0| + |u_1| + |\xi|). \tag{127}$$

For completeness, a crude explicit choice is available. Put

$$D := (2+b)(3+2a)+1, \qquad C := \left(1+\frac{2(K-1)}{\sqrt{b}}\right) D^{K-1} + 3.$$

Indeed, with $M_k = \max\{|u_k|, |u_{k-1}|, |\xi|\}$, the finite-step recurrence gives $M_{k+1} \le DM_k$, because $|q_k| \le 2+b$, $|\beta_k| \le 1+a$, and $s\lambda < 1$. Combine $M_K \le D^{K-1}M_1$ and $|v_K| \le 2M_K$ with (126). This proves (127) uniformly even for arbitrarily small positive λ.

Positive spectral values

Fix $\lambda > 0$. Then $q_k \to q_\infty := 1 - s\lambda \in (0,1)$. Relative to the two weights of J_k, the coefficients in (122) are

$$q_k \quad \text{and} \quad R_k(\lambda) := \frac{q_k^2 \beta_k^2 t_{k-1}}{t_k q_{k-1}},$$

and both tend to q_∞. Choose $\vartheta \in (q_\infty, 1)$. On a sufficiently late tail,

$$\sqrt{J_{k+1}} \le \sqrt{\vartheta}\big(\sqrt{J_k} + |z_k - z_{k-1}|\big).$$

The additive term tends to zero. A geometric contraction comparison gives $J_k \to 0$, and $z_k \to \xi$ gives $u_k \to \xi$. This pointwise argument is valid for every $a > 0$; its starting index may depend on λ. It is (127), not this pointwise argument, that will justify spectral dominated convergence.

The flat recurrence

At $\lambda = 0$, put $w_k = k(u_k - u_{k-1})$ and $Z_k = (u_k, w_k)^{\mathsf{T}}$. Directly from (120),

$$Z_{k+1} = T_k Z_k, \qquad T_k = \begin{pmatrix} 1 - b/k^2 & (1-a/k)(1-b/k^2)/k \\ -b(k+1)/k^2 & (1+1/k)(1-a/k)(1-b/k^2) \end{pmatrix}. \tag{128}$$

Hence $T_k = I + B_0/k + O(k^{-2})$, where

$$B_0 = \begin{pmatrix} 0 & 1 \\ -b & 1-a \end{pmatrix}.$$

For $a \neq 1$, let $\kappa = |a-1|$, $\sigma = \operatorname{sign}(a-1)$, and define

$$P = \begin{pmatrix} \dfrac{\kappa^2 + b(b+1)}{2b\kappa} & \dfrac{\sigma}{2b} \\ \dfrac{\sigma}{2b} & \dfrac{b+1}{2b\kappa} \end{pmatrix}. \tag{129}$$

Its first diagonal entry is positive and

$$\det P = \frac{\kappa^2 + (b+1)^2}{4b\kappa^2} > 0, \qquad B_0^{\mathsf{T}} P + P B_0 = -\sigma I.$$

Thus $\mathcal{E}_k = Z_k^{\mathsf{T}} P Z_k$ satisfies

$$\mathcal{E}_{k+1} - \mathcal{E}_k = -\frac{\sigma}{k} \|Z_k\|^2 + O(k^{-2}) \|Z_k\|^2 . \tag{130}$$

For $a > 1$, positive definiteness gives an eventual contraction by $1 - c/k$ for some $c > 0$. Therefore $Z_k \to 0$.

For $0 < a < 1$, every nonzero sufficiently late state instead satisfies an eventual lower growth factor $1 + c/k$, so $\mathcal{E}_k \to \infty$. We must check that u_k itself is unbounded. The exact second-component recurrence is

$$w_{k+1} = \alpha_k w_k - \delta_k u_k, \quad \alpha_k = (1+1/k)(1-a/k)(1-b/k^2) \to 1, \quad \delta_k = \frac{b(k+1)}{k^2} \to 0. \tag{131}$$

If u_k were bounded, energy divergence would imply $|w_k| \to \infty$. Choose a late index at which $|w_k| \geq 1$, $\alpha_k \geq 1/2$, and $|\delta_k u_k| < 1/4$, with these inequalities valid thereafter. Equation (131) then prevents a change of sign of w_k. But

$$u_n - u_N = \sum_{j=N+1}^{n} \frac{w_j}{j}$$

is unbounded by the harmonic series, a contradiction.

When $a = 1$, take $P = \operatorname{diag}(b, 1)$. Now $B_0^{\mathsf{T}} P + P B_0 = 0$, so

$$|\mathcal{E}_{k+1} - \mathcal{E}_k| \leq \frac{C}{k^2} \mathcal{E}_k$$

on a tail. For a nonzero tail, the products of $1 \pm C/k^2$ bound $\mathcal{E}_k$ above and away from zero. If $u_k \to 0$, then $|w_k|$ is eventually bounded below by a positive constant. The same sign and harmonic-sum argument using (131) is a contradiction. Thus a nonzero flat tail does not converge to zero at $a = 1$.

Survival of the zero-velocity initialization

For the instability assertion we need specifically $u_0 = u_1 = 1$. We verify that this initial state has a nonzero tail for every $b > 0$ and $0 < a \leq 1$.

The determinant of (128) is

$$(1 + 1/k)(1 - a/k)(1 - b/k^2).$$

If b is not an integer square and $a < 1$, all transitions are invertible. If b is not an integer square and $a = 1$, the first momentum factor vanishes, but its output has $(u_2, w_2) = (1 - b, -2b) \neq (0, 0)$; later transitions are invertible.

Suppose $b = m^2$ for an integer $m \geq 1$. For $m = 1$, $u_m = u_1 = 1$. For $m \geq 2$, the first update gives $u_2 = 1 - m^2 < 0$. If $2 \leq j < m$ and u_j, u_{j-1} have opposite nonzero signs, then

$$(2 - a/j)u_j - (1 - a/j)u_{j-1}$$

has the nonzero sign of u_j, and multiplication by $1 - m^2/j^2 < 0$ reverses it. Induction yields $u_m \neq 0$. At the singular step,

$$u_{m+1} = 0, \qquad w_{m+1} = -(m+1)u_m \neq 0.$$

Every later transition is invertible. This proves survival in all cases. In particular, the prescribed scalar sequence is unbounded for $0 < a < 1$, and does not tend to zero for $a = 1$.

Spectral dominated convergence

For each finite k, the affine recurrence gives polynomials A_k, B_k, C_k such that

$$x_k = A_k(Q)x_0 + B_k(Q)x_1 + C_k(Q)x^*. \tag{132}$$

They are the scalar responses to the three initial-data triples. Applying (127) to each triple gives

$$|A_k(\lambda)|, |B_k(\lambda)|, |C_k(\lambda)| \leq C, \qquad \lambda \in [0, L], \quad k \geq 0.$$

If $a > 1$, the positive-mode and flat-mode arguments show $A_k(\lambda), B_k(\lambda) \to 0$ for every $\lambda \geq 0$. Also $C_k(\lambda) \to 1$ for $\lambda > 0$, whereas $C_k(0) = 0$ for all k.

Let E_Q be the projection-valued spectral measure of Q. For instance,

$$\|A_k(Q)x_0\|^2 = \int |A_k(\lambda)|^2 \, d\langle E_Q(\lambda)x_0, x_0\rangle \longrightarrow 0$$

by dominated convergence against a finite scalar measure. The same holds for $B_k(Q)x_1$. Finally,

$$\|(C_k(Q) - I)x^*\|^2 = \int |C_k(\lambda) - 1|^2 \, d\langle E_Q(\lambda)x^*, x^*\rangle \longrightarrow 0,$$

because the zero spectral mass is $\|P_{\ker Q}x^*\|^2 = 0$. This proves the assertion for $a > 1$ without a spectral gap.

At $a = 1$, the same domination remains valid and every positive mode has the same limit. If $\ker Q = \{0\}$, all relevant spectral measures have no atom at zero, so the argument still proves strong convergence. In general, $\beta_1 = 0$ makes every iterate from x_2 onward independent of x_0. The kernel component is

$$P_{\ker Q}x_k = p_k(0)P_{\ker Q}x_1, \qquad k \geq 1,$$

where $p_0(0) = p_1(0) = 1$ is the scalar sequence just considered. Its nonconvergence to zero proves the necessity of $P_{\ker Q}x_1 = 0$. If this projection vanishes, dominated convergence on the positive spectral part proves sufficiency.

Coercivity and the subcritical obstruction

Suppose $Q \geq \mu I$ for some $\mu > 0$. For every $a > 0$, the coefficients q_k and $R_k(\lambda)$ used in the positive-mode argument converge uniformly on $[\mu, L]$ to $1 - s\lambda \leq 1 - s\mu < 1$. Their associated norms have coefficients bounded uniformly above and away from zero. The moving centers converge uniformly to ξ, and their increments tend uniformly to zero. The geometric contraction comparison therefore holds uniformly on the spectrum. Bounding the finite initial segment and applying (132) proves convergence for every target and initialization. The constants here may depend on μ.

Conversely, suppose $0 < a < 1$ and $0 \in \sigma(Q)$. Define the homogeneous response polynomials by $p_0(\lambda) = p_1(\lambda) = 1$ and

$$p_{k+1}(\lambda) = \left(1 - s\lambda - \frac{b}{k^2}\right)\left[p_k(\lambda) + \left(1 - \frac{a}{k}\right)(p_k(\lambda) - p_{k-1}(\lambda))\right].$$

The flat analysis proves $\sup_k |p_k(0)| = \infty$ for this prescribed initialization. Continuous functional calculus gives

$$\|p_k(Q)\| = \max_{\lambda \in \sigma(Q)} |p_k(\lambda)| \geq |p_k(0)|.$$

The family of bounded linear operators $p_k(Q)$ is therefore not uniformly bounded. By the uniform boundedness principle, there exists $v \in \mathcal{H}$ such that

$$\sup_k \|p_k(Q)v\| = \infty.$$

For $x^* = 0$ and $x_0 = x_1 = v$, (119) gives exactly $x_k = p_k(Q)v$. This is the claimed unbounded orbit with zero initial velocity. The argument uses a zero spectral value, not a zero eigenvector, so it applies to injective noncoercive operators.

For a bounded positive-semidefinite selfadjoint operator, $0 \notin \sigma(Q)$ is equivalent to coercivity. The preceding arguments prove all necessity and sufficiency statements in Theorem 6. ∎

Remark. For a concrete setting, take $\mathcal{H} = \ell^2$ and $Qe_n = n^{-1}e_n$. This bounded operator is injective and noncoercive. Its zero-target quadratic has a unique minimizer. Every orbit converges when $a = 1$, while an unbounded zero-velocity orbit exists for each $0 < a < 1$. Pointwise convergence at every fixed positive spectral value is compatible with the latter failure; a uniform spectral bound is the missing ingredient.

8 Comparison of the critical assumptions

The parameter regime of Theorem 1 contains the displayed critical regime in [1, Theorem 3.3]. Indeed, two of the latter's conditions are

$$\frac{1}{\sqrt{sc}} < \frac{2+q}{1+q}\lambda - \delta, \qquad \frac{1}{\sqrt{sc}} < \delta - \lambda,$$

where $q > 0$ is an auxiliary parameter. With the notation in (2) and $\ell = \lambda\sqrt{sc}$, they imply

$$\frac{(a+1)(1+q)}{2+q} < \ell < a - 1.$$

It follows that $a > 3 + 2q > 3$. The accompanying step condition in [1] gives $sL < q/(q+1) < 1$. Theorem 1 therefore establishes selection throughout that critical regime without its other auxiliary parameter restrictions.

We next compare the geometric condition. In the present setting, the critical Attouch–Czarnecki condition in [13, Assumption 2] asks for finiteness of

$$\sum_{k=1}^{\infty} k\left[(f - f_\star)^*(\varepsilon_k p) - \sigma_S(\varepsilon_k p)\right] \tag{133}$$

for the relevant normal vectors p, where σ_S is the support function and the superscript $*$ denotes the Fenchel conjugate. For the norm upper objective, the vector $p = -x^*$ is such a normal vector. The following example shows that the absence of (133) from Theorem 1 is a genuine difference in hypotheses.

Proposition 8.1. There is a convex continuously differentiable function on $\mathbb{R}^2$ with globally Lipschitz continuous gradient, a nonaffine minimizer set, and a nonzero minimum norm minimizer x^*, such that (133) diverges for $p = -x^*$ and every schedule $\varepsilon_k = c/(k+\beta)^2$, with $c > 0$ and $\beta \geq 0$. Moreover, it has bounded sublevels, satisfies (8) globally with $\gamma = 2$, and satisfies Theorem 4(ii) for every finite $\gamma > 1$. Thus every $a > 1$ is permitted by that theorem for this fixed objective.

Proof. Let $e_1 = (1, 0)$ and let C be the closed unit ball centered at $2e_1$. Then

$$P_C(0) = e_1, \qquad \sigma_C(q) = 2\langle q, e_1\rangle + \|q\|, \qquad -e_1 \in N_C(e_1).$$

Define the increasing convex function $\psi : [0, \infty) \to [0, \infty)$ by

$$\psi(r) = \begin{cases} \displaystyle\int_0^r e^{-1/t}\, dt, & 0 \leq r \leq 1, \\ \psi(1) + \dfrac{e^{-1}}{2}(r^2 - 1), & r \geq 1. \end{cases} \tag{134}$$

The integrand at 0 is assigned the value 0. Set

$$\Phi(z) := \psi(\|z\|), \qquad f(x) := \psi(\operatorname{dist}(x, C)).$$

The function f is convex because distance to a closed convex set is convex and ψ is convex and nondecreasing. Its minimizer set is exactly C.

For $0 < r \leq 1$,

$$\psi'(r) = e^{-1/r}, \qquad \psi''(r) = r^{-2}e^{-1/r} \leq 4e^{-2}, \qquad \frac{\psi'(r)}{r} \leq e^{-1}.$$

For $r \geq 1$, both $\psi''(r)$ and $\psi'(r)/r$ equal e^{-1}. The radial gradient $\nabla\Phi$ is therefore globally $4e^{-2}$-Lipschitz, including across $r = 0$ and $r = 1$. Moreover,

$$\nabla f(x) = \nabla\Phi(x - P_C x).$$

This formula is valid on the boundary of C because $\psi'(0) = 0$. Since $I - P_C$ is nonexpansive, ∇f is globally $4e^{-2}$-Lipschitz. In particular, f satisfies the smoothness assumptions of Theorem 1.

The identity $f(x) = \inf_{z\in C} \Phi(x - z)$ gives

$$f^*(q) = \sigma_C(q) + \psi^*(\|q\|), \qquad \psi^*(t) := \sup_{r\geq 0}\{tr - \psi(r)\}. \tag{135}$$

For $0 < t < e^{-1}$, the maximizing radius is $r_t = 1/\log(1/t)$. For $0 < r \leq 1$, substituting $u = 1/t$ in (134) gives

$$\psi(r) = \int_{1/r}^{\infty} \frac{e^{-u}}{u^2}\, du \leq r^2 e^{-1/r}.$$

Thus, for $0 < t \leq e^{-2}$,

$$\frac{t}{2\log(1/t)} \leq \psi^*(t) \leq \frac{t}{\log(1/t)}. \tag{136}$$

Since $f_\star = \min f = 0$, taking $p = -e_1$ in (135) yields

$$(f - f_\star)^*(\varepsilon_k p) - \sigma_C(\varepsilon_k p) = \psi^*(\varepsilon_k).$$

For $\varepsilon_k = c/(k + \beta)^2$, the lower bound in (136) shows that $k\psi^*(\varepsilon_k)$ is bounded below, for all sufficiently large k, by a positive constant times $1/(k \log k)$. Its sum diverges. Since $x^* = e_1$, this is the required normal vector $p = -x^*$.

To verify relative flatness, let $z = P_C x$ and $r = \operatorname{dist}(x, C) > 0$. Projection normality gives

$$\langle \nabla f(x), x - x^*\rangle = \frac{\psi'(r)}{r}\big(r^2 + \langle x - z, z - x^*\rangle\big) \geq r\psi'(r).$$

For $0 < r \leq 1$, the derivative of $r\psi'(r) - 2\psi(r)$ is $e^{-1/r}(1/r - 1) \geq 0$, and its limit at zero is zero. For $r \geq 1$, this difference is the nonnegative constant $e^{-1} - 2\psi(1)$. Thus $r\psi'(r) \geq 2\psi(r)$ globally. Inside C both sides vanish, proving (8) with $\gamma = 2$.

The function ψ tends to infinity quadratically and C is bounded, so every sublevel of f is bounded. Fix any finite $\gamma > 1$ and put $\Delta = \psi(1/\gamma) > 0$. If $f(x) \leq \Delta$, then $r \leq 1/\gamma < 1$. The estimate $\psi(r) \leq r^2 e^{-1/r}$ gives

$$r\psi'(r) \geq \frac{\psi(r)}{r} \geq \gamma\psi(r).$$

Projection normality gives the required flatness on this entire low sublevel. For any chosen $a > 1$, take a finite $\gamma > \max\{1, 2/(a - 1)\}$ in Theorem 4(ii). The objective and regularization center remain fixed; only the admissible exponent and entry sublevel depend on a. ∎

An interior minimizer set without target flatness

The preceding example is covered by the relative-flatness theorem. We next give one that satisfies neither target-relative flatness nor targeted summability, while retaining the hypotheses of Theorem 5.

Proposition 8.2. There is a convex C^1 function $f : \mathbb{R}^2 \to \mathbb{R}$ with globally 1-Lipschitz gradient and bounded sublevels whose minimizer set is the closed unit ball centered at $2e_1$. Its minimum norm point is $x^* = e_1$. For every $\gamma > 1$ and $\Delta > 0$, some x with $0 < f(x) \le \Delta$ satisfies

$$\langle \nabla f(x), x - x^* \rangle < \gamma f(x).$$

Its targeted series (7) diverges for every positive scale. Nevertheless, for every $a > 1$, $b > 0$, and $0 < s < 1$, every orbit of (1) satisfies (4) for this fixed objective.

Proof. Set $r_1 = 1/4$, $r_{j+1} = r_j^2$, and $\lambda_j = e^{-1/r_j}$ for $j \ge 1$. Define a continuous function $\chi : [0, \infty) \to [0, \infty)$ by $\chi(0) = 0$ and

$$\chi(r) = \begin{cases} \lambda_j, & 2r_{j+1} \le r \le r_j, \quad j \ge 1, \\ \lambda_{j+1} + \dfrac{\lambda_j - \lambda_{j+1}}{r_{j+1}}(r - r_{j+1}), & r_{j+1} \le r \le 2r_{j+1}, \quad j \ge 1, \\ \lambda_1 + r - r_1, & r \ge r_1. \end{cases} \tag{137}$$

The intervals cover $(0, \infty)$ and their endpoint values agree. Every connecting slope is nonnegative and at most

$$\frac{e^{-1/r_j}}{r_j^2} \le 16e^{-4} < 1.$$

Indeed, $r \mapsto r^{-2}e^{-1/r}$ is increasing on $(0, 1/4]$. The outer slope is one. Thus χ is nondecreasing and globally 1-Lipschitz, with $0 \le \chi(r) \le r$ and $\chi(r) > 0$ for $r > 0$.

Put

$$\psi(r) := \int_0^r \chi(t)\, dt, \qquad C := \{x \in \mathbb{R}^2 : \|x - 2e_1\| \le 1\}, \qquad f(x) := \psi(\operatorname{dist}(x, C)).$$

Convexity follows because ψ is convex and nondecreasing. The function is C^1, with gradient zero on C and

$$\nabla f(x) = \chi(R - 1)\frac{x - 2e_1}{R}, \qquad R = \|x - 2e_1\| > 1,$$

outside C. To check the global Lipschitz constant directly, set $\omega(R) = \chi((R-1)_+)$, which is 1-Lipschitz and satisfies $0 \le \omega(R) \le R$. For unit vectors u, v and $R, T > 0$,

$$\begin{aligned} \|\omega(R)u - \omega(T)v\|^2 &= (\omega(R) - \omega(T))^2 + \omega(R)\omega(T)\,\|u - v\|^2 \\ &\le (R - T)^2 + RT\,\|u - v\|^2 = \|Ru - Tv\|^2. \end{aligned}$$

Continuity covers zero radii. Hence ∇f is globally 1-Lipschitz. The outer affine branch of χ makes ψ grow quadratically, so sublevels are bounded. Strict positivity of $\psi(r)$ for $r > 0$ gives $S = C$ and $x^* = e_1$.

At $x_j = (1 - r_j)e_1$,

$$\langle \nabla f(x_j), x_j - x^* \rangle = r_j \lambda_j, \qquad f(x_j) = \psi(r_j) \geq \lambda_j (r_j - 2r_{j+1}).$$

Therefore

$$1 \leq \frac{\langle \nabla f(x_j), x_j - x^* \rangle}{f(x_j)} \leq \frac{1}{1 - 2r_j} \longrightarrow 1,$$

while $f(x_j) > 0$ and $f(x_j) \to 0$. This proves failure of every stated target-flatness condition.

As in (135), radial maximization gives $f^*(q) = \sigma_C(q) + \psi^*(\|q\|)$. For $p = -e_1$, the targeted defect is therefore $D(t) = \psi^*(t)$, finite for every $t \geq 0$ because of the quadratic outer growth. Since $\psi(r_{j+1}) \leq \lambda_{j+1} r_{j+1}$, testing the conjugate at r_{j+1} gives

$$D(t) \geq \frac{1}{2} t r_{j+1}, \qquad 2\lambda_{j+1} \leq t \leq \lambda_j.$$

These nonempty intervals are pairwise disjoint, and

$$\begin{aligned} \int_{2\lambda_{j+1}}^{\lambda_j} \frac{D(t)}{t^2}\, dt &\geq \frac{r_{j+1}}{2} \log \frac{\lambda_j}{2\lambda_{j+1}} \\ &= \frac{1}{2}(1 - r_j - r_j^2 \log 2) \geq \frac{11}{32}. \end{aligned}$$

Thus $\int_0^\tau D(t) t^{-2}\, dt = \infty$ for every $\tau > 0$. Lemma 1 proves divergence of the targeted series at every positive scale. Finally, C has nonempty ambient interior, so Theorem 5 supplies all claimed convergence conclusions for this single function. ∎

Coupled examples with an empty-interior minimizer set

Use the exponential staircase primitive ψ from Proposition 8.2, and put $g(w) = \psi(\operatorname{dist}(w, [1, 3]))$. A simple subquartic example is

$$f_3(u, n) = g(u + n) + H_3(n), \qquad H_3(n) = \begin{cases} |n|^3/3, & |n| \leq 1, \\ |n|^2 - |n| + 1/3, & |n| \geq 1. \end{cases}$$

Its gradient is globally 4-Lipschitz, its sublevels are bounded, and $S = [1, 3] \times \{0\}$. On $f_3 \leq 1/3$ it satisfies $f_3 \geq \operatorname{dist}(\,\cdot\,, \operatorname{aff} S)^3/3$. On the line $u = 2 - n$, however, $f_3 = |n|^3/3$ near $n = 0$, so no transverse quadratic lower bound holds. At $(1 - r_j, 0)$ the target-flatness ratio is the same as in Proposition 8.2. The substitution $w = u + n$ gives

$$f_3^*(\alpha, \beta) = g^*(\alpha) + H_3^*(\beta - \alpha),$$

so its targeted defect is $\psi^*(t) + H_3^*(t)$ and its critical AC series diverges. Theorem 5(i) nevertheless applies for every $a > 1$, $b > 0$, and $0 < s < 1/4$.

At quartic growth we can make ordinary relative flatness fail at *every* minimizer, including all possible relative-interior comparison centers. The affine hull can also be shifted away from the regularization anchor.

Proposition 8.3. There is a convex C^1 function $f : \mathbb{R}^2 \to \mathbb{R}$ with globally 3-Lipschitz gradient and bounded sublevels such that

$$S = [1,3] \times \{1\}, \qquad A = \operatorname{aff} S = \mathbb{R} \times \{1\}, \qquad x^* = (1,1),$$

and

$$f(x) \geq \frac{1}{512} \operatorname{dist}(x,A)^4 \qquad \text{whenever } f(x) \leq \frac{1}{1024}. \tag{138}$$

No positive transverse growth bound of order less than four holds on any positive low sublevel. For every $z \in S$, every $\gamma > 1$, and every $\Delta > 0$, some x with $0 < f(x) \leq \Delta$ satisfies $\langle \nabla f(x), x - z \rangle < \gamma f(x)$. The targeted AC series diverges for every positive scale. On every positive low sublevel the objective admits no orthogonal separation into two scalar functions. Nevertheless, at $a = 3$, every $b > 0$ and $0 < s < 1/3$ give all conclusions in (4) for arbitrary initial pairs.

Proof. Keep $r_1 = 1/4$, $r_{j+1} = r_j^2$, and set $\nu_j = r_j^3$. Define $\eta(0) = 0$ and

$$\eta(r) = \begin{cases} \nu_j, & 2r_{j+1} \leq r \leq r_j, \\ \nu_{j+1} + \dfrac{\nu_j - \nu_{j+1}}{r_{j+1}}(r - r_{j+1}), & r_{j+1} \leq r \leq 2r_{j+1}, \\ \nu_1 + r - r_1, & r \geq r_1. \end{cases}$$

The first two cases range over $j \geq 1$. The connecting slopes equal $r_j - r_j^4 \in (0, 1/4]$. Hence η is continuous, nondecreasing and globally 1-Lipschitz, with $0 \leq \eta(r) \leq r$ and $\eta(r) > 0$ for $r > 0$. Set

$$\Phi(r) = \int_0^r \eta(t)\, dt, \qquad N(v) = \Phi(|v|), \qquad f(u,n) = g(u + n - 1) + N(n - 1).$$

The derivative of N is globally 1-Lipschitz; for arguments of opposite signs use $\eta(r) \leq r$. The ridge $g(u + n - 1)$ has gradient Lipschitz constant at most two. Thus f is convex, C^1, and globally 3-smooth. Quadratic outer growth of both scalar profiles bounds all sublevels. Their zero sets give the stated S, A, x^*.

For $r_{j+1} \leq r \leq r_j$, write $R = r_{j+1}$. If $r \geq 4R$, integration of the upper plateau gives

$$\Phi(r) \geq \nu_j(r - 2R) \geq \frac{1}{2}\nu_j r \geq \frac{1}{2} r^4.$$

If $R \leq r \leq 4R$, the interval $[R/2, R]$ lies in the next plateau $[2R^2, R]$. Thus

$$\Phi(r) \geq \Phi(R) \geq \frac{1}{2} R^4 \geq \frac{1}{512} r^4.$$

These cases prove the lower bound on $0 \le r \le 1/4$. Also $\Phi(1/4) \ge \nu_1(r_1 - 2r_2) = 1/512$. Consequently $f \le 1/1024$ forces $|n-1| < 1/4$, proving (138).

Fix any $z = (z_u, 1) \in S$ and take $x_j = (z_u - r_j, 1 + r_j)$. Its shear coordinate equals z_u, where $g = g' = 0$. Therefore

$$f(x_j) = \Phi(r_j), \qquad \nabla f(x_j) = (0, \nu_j), \qquad \langle \nabla f(x_j), x_j - z \rangle = r_j^4.$$

Since $r_j^4(1 - 2r_j) \le \Phi(r_j) \le r_j^4$, the relative-flatness ratio lies between 1 and $(1-2r_j)^{-1}$ and tends to one. The same points satisfy $f(x_j)/\operatorname{dist}(x_j, A)^q \le r_j^{4-q} \to 0$ for every $q < 4$. There is also no full-set lower growth bound of any finite positive order: at $(1 - r_j, 1)$ the value is at most $r_j e^{-1/r_j}$ and the distance to S is r_j.

For a dual vector (α, β), the substitution $v = n - 1$, $w = u + n - 1$ yields

$$f^*(\alpha, \beta) = g^*(\alpha) + N^*(\beta - \alpha) + \beta.$$

All conjugates are finite at finite arguments because of quadratic outer growth. At the target normal $p = (-1, -1)$, use $g^*(-t) = -t + \psi^*(t)$ and $N^*(0) = 0$ to obtain

$$f^*(tp) = -2t + \psi^*(t), \qquad \sigma_S(tp) = -2t, \qquad D(t) = \psi^*(t).$$

The divergence proved in Proposition 8.2 is therefore unchanged.

Finally choose a positive normal displacement inside a connecting interval of η, where $N'' = \xi > 0$. At that same displacement, choose one shear coordinate in $(1, 3)$ and another inside a connecting interval of the exponential profile just outside $[1, 3]$, where $g'' = \tau > 0$. The two Hessians are

$$B = \begin{pmatrix} 0 & 0 \\ 0 & \xi \end{pmatrix}, \qquad C = \tau \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + B.$$

They do not commute. Both points lie in arbitrarily low sublevels when the connecting intervals approach zero. Any orthogonal separation into two scalar functions would diagonalize all these Hessians in the same coordinates, a contradiction. The affine shear separates the formula but changes the Euclidean gradient metric, so it does not separate the recurrence.

Theorem 5(ii) applies with the explicit constants in (138), giving the asserted convergence and rates. ∎

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